

Persianate Geomancy from Ṭūsī to the Millennium

A Preliminary Survey

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Abstract

The divinatory art of geomancy (*‘ilm al-raml*), Arabic answer to the *I Ching*, was third in popularity only to astrology and oneiromancy throughout the premodern Islamo-Christianate world, and is still widely practiced today from Iran and Yemen to West Africa and Madagascar. Despite its ubiquity among scholarly and ruling elites, however, the reflexive scholarly elision of occultism from Islamicate intellectual and cultural history means that this science has been very poorly studied beyond its reception by Europe. This holds especially true for the early modern Persianate world: in the wake of the occultist renaissance that swept the Islamicate heartlands from the late 8th/14th century onward, geomancers emerged as influential fixtures at Mughal, Safavid and Ottoman courts, and geomancy, offering detailed knowledge of past, present and future, was widely lionized as a means for controlling history – yet the subject remains almost untouched. And despite its status as a mathematical occult science predicated on a neoplatonic-neopythagorean system and animated by the twin principles of correspondence and secondary causation, its importance to the history of science has likewise only been recognized in its European context. Extending the work of Emilie Savage-Smith, Marion B. Smith and others, I therefore provide the first survey, necessarily tentative, of the development of geomantic theory and practice in the persophone world from the 7th/13th century onward, when Naṣīr al-Dīn Ṭūsī (d. 672/1274) inaugurated a specifically Persianate geomantic tradition for the benefit of his Mongol patron Hülegü. With Ṭūsī as precedent, geomancy went on to exercise some of the best minds of the early modern Persianate world as a mainstream occult-scientific tradition, and enjoyed a crescendo of popularity in the 10th/16th century with the approach of the Islamic millennium. After sketching the development of the eastern geomantic tradition, then, I identify the philosophical-scientific reasons for its great vitality and cultural prestige down to the present, with particular attention to authors on geomancy like Luṭf Allāh Nishāpūri Samarqandi (fl. 812/1409), Sharaf al-Dīn Yazdī (d. 858/1454), Shams al-Dīn Khafī (d. 942/1535) and Hidāyat Allāh Munajjim-i Shīrāzī (fl. 1001/1593), court geomancer to Emperor Akbar at the turn of the millennium.

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This art [of geomancy] is much practiced in all civilized lands.
–Ibn Khaldūn¹

Caput Draconis!
–Gryffindor password²

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¹ *Muqaddima*, Darwish al-Juwaydi, ed., Beirut: al-Maktaba al-‘Aṣriyya, 1995, 108.

Geomancy, the ‘science of sand’ (*‘ilm al-raml*), is a uniquely Arabic and Islami-cate occult science³ that captured the intellects and imaginations of scholarly elites and their royal patrons throughout the premodern Islamo-Christianate world. As a complex divinatory science based on a binary code, geomancy stands precise cognate to the ancient Chinese *I Ching*, but commanded in its prime a far greater territorial spread: it was – and in many cases still is – regularly practiced as a single tradition from Fez, Paris and Timbuktu to Kashgar, Kabul and Delhi, and calved simpler versions throughout sub-Saharan Africa and thence the western hemisphere that remain very much in use. Throughout its Afro-Eurasian domain, geomancy was third in popularity only to astrology and oneiromancy; indeed, due to its early incorporation of astrological correspondences, it was often considered a form of “terrestrial astrology” requiring much less astronomical expertise while remaining richly informative.⁴ This association is exemplified by the fact that in the early modern Persianate east individuals designated *munajjims*, or court astronomer-astrologers, increasingly acted as court geomancers as well.⁵ *Raml* and *jafr*, or letter divination, were also often seen as a complementary pair of divinatory techniques based on number theory (*‘ilm al-‘adad*),⁶ the

² J. K. Rowling, *Harry Potter and the Sorcerer’s Stone*, New York: Scholastic, 1998, 130. The Latin names of other geomantic figures are similarly utilized in the Harry Potter series, including Albus, Rubeus and Fortuna Major.

³ The standard Arabic term for the occult sciences, including astrology (*ahkām al-nujūm*), alchemy (*kimiyā*) and a variety of magical and divinatory techniques, is *‘ulūm ghariba*, meaning those sciences that are unusual, rare or difficult, i.e., elite; less frequently used terms are *‘ulūm khafiyya* and *‘ulūm ghāmiḍa*, sciences that are hidden or occult. These terms are routinely used in classifications of the sciences, biographical dictionaries, chronicles, etc. Its 19th-century European flavor notwithstanding, the term occultism is therefore used here simply to denote a scholarly preoccupation with one or more of the occult sciences as discrete natural-philosophical or mathematical disciplines. *Occultism* is thus to be strictly distinguished from *mysticism* and *esotericism*, for all that scholars from Corbin onward have habitually and perniciously disappeared the former into the latter; I discuss this disappearance in the final section of this article. More generally, ‘science’ and ‘scientific’ awkwardly translate the Arabic terms *‘ilm* (lit., ‘knowledge’) and *‘ilmi*, which rather correspond to the broader concept of *scientia* or *Wissenschaft*, including *Geisteswissenschaft*. See Raymond Williams, “Science”, in: idem. *Keywords: A Vocabulary of Culture and Society*, New York: Oxford Univ. Press, 1983, 276–280.

⁴ See e.g. Ibn Khaldūn, *Muqaddima*, 107, 108. Significantly, in his classification of the sciences Ḥājī Khalifa (d. 1067/1657) categorizes geomancy under the rubric of judicial astrology (Toufic Fahd, *La divination arabe: Études religieuses, sociologiques et folkloriques sur le milieu natif de l’Islam*, Leiden: Brill, 1966, 41).

⁵ Fahd, *La divination arabe*, 200. This blending of geomancy and astrology began early on; one of the first codifiers of the Arabic geomantic tradition, Ibn Maḥfūf (d. before 664/1265), is typically called *al-Munajjim*; Fahd, *La divination arabe*, 201.

⁶ That is, the ‘science of number,’ or arithmetic, particularly as associated with Pythagoras and Nicomachus. The 4th/10th-century group of neopythagorean-occultist encyclopedists known as the Brethren of Purity, for instance, vigorously assert arithmetic to be the basis for all other mathematical, natural, psychological and metaphysical sciences, an ascending epistemological series that culminates in sciences like astrology, alchemy and magic; see

first employing dots (*niqāt*) and figures (*ashkāl*) and the second the letters (*ḥurūf*) of the Arabic alphabet (also derived from a dot), and both equally productive of knowledge of past, present and future.

Knowledge of geomancy entered the Christianate world in the 6th/12th century with Latin translations of Arabic works, its popularity burgeoning from there.⁷ In the Islamicate world, geomancy was typically associated in the first place with the prophets Idris (Enoch or Hermes)⁸ and Daniel and the Indian sage Ṭumṭum, not to mention a number of other standard occultist authorities.⁹ Intellectual genealogies of the science in Arabic and Persian works on the subject thus presuppose a pre-Islamic Near Eastern or Indian origin, as well as an early Berber connection; the otherwise unknown Abū ‘Abd Allāh Muḥammad al-Zanātī (fl. before 629/1230), presumably of the Berber Zanāta tribe, is acclaimed as the first major Arabic exponent of geomancy.¹⁰

While geomancy fell out of mainstream use in post-Enlightenment Europe,¹¹ it experienced no such decline in the “un-Enlightened” Islamicate world, and partic-

e.g. Nader El-Bizri, “Epistolary Prolegomena: On Arithmetic and Geometry”, in: *Epistles of the Brethren of Purity: The Ikhwān al-Ṣafā’ and Their Rasā’il: An Introduction*, Nader El-Bizri, ed., Oxford: Oxford Univ. Press, 2008, 180–213.

⁷ The extant Latin treatises are surveyed in Thérèse Charmasson, *Recherches sur une technique divinatoire: la géomancie dans l’occident médiéval*, Paris: Champion, 1980. After Hugo of Santalla’s (fl. 1145) pathbreaking translations of two treatises from the Arabic under the titles *Ars geomantiae* and *Geomantia nova*, notable examples include Heinrich Cornelius Agrippa von Nettesheim’s (d. 1535) *In Geomanticam disciplinam lectura* (in: *Opera*, 12 vols., Hildesheim: Olms, 1970, vol. 1, 500–526 [repr. Basel 1578]), Christophe de Cattan’s (fl. 1558) *La Géomance* (subtitled *Livre non moins plaisant & recreatif, que d’ingenieuse invention, pour scavoir toutes choses, presentes, passées, & à advenir, Avec la Roüe de Pythagoras*, Paris: Gilles, 1558; English translation by Francis Sparry in 1591) and Robert Fludd’s (d. 1637) *De geomantia* (in: *Fasciculus geomanticus, in quo varia variorum opera geomantica continentur*, Veronae, 1687, 1–170) (Marion B. Smith, “The Nature of Islamic Geomancy with a Critique of a Structuralist’s Approach”, in: *Studia Islamica* 49 (1979), 5–38, here: 8; Stephen Skinner, *Geomancy in Theory and Practice*, 2nd ed., St. Paul, Minn.: Golden Hoard Press, 2011, 112–116).

⁸ In his *Nafahāt al-Asrār* (on which see below), Muḥammad Bāqir Ṭabāṭabā’i Yazdī invokes a statement by Imam ‘Alī attributing the science to Idris as received from Gabriel (lithograph Najaf: Dār al-Kutub al-‘Irāqīyya, 1359/1940, 10).

⁹ These include Imams ‘Alī b. Abi Ṭālib and Ja‘far al-Ṣādiq as well as a number of other biblical prophets. On Ṭumṭum al-Hindi see A. Hauber, “Tomṭom (Ṭimṭim) = Dandamiz = Dindymus?”, in: *Zeitschrift der Deutschen Morgenländischen Gesellschaft* 63 (1909), 457–472.

¹⁰ The best study to date of the Arabic geomantic tradition through the 7th/13th century is Emilie Savage-Smith and Marion B. Smith, “Islamic Geomancy and a Thirteenth-Century Divinatory Device: Another Look”, in: *Magic and Divination in Early Islam*, Emilie Savage-Smith, ed., Aldershot: Ashgate, 2004, 211–276, which expands upon their findings in a similarly titled publication from 1980; in particular, they show that it was fully developed and established by the middle of that century.

¹¹ Geomancy was largely replaced by the *I Ching* as the divinatory art of choice in modern Euro-American occultism, particularly as represented by the Hermetic Order of the Golden Dawn; this Chinese cognate was also the subject of considerable interest to C. G. Jung (along with a number of other occult sciences), who postulated that number regulates both psyche and matter and described divination as the ‘science of synchronicity’ (see e.g. *Jung on Synchronicity and the Paranormal*, Roderick Main, ed., Princeton: Princeton Univ. Press,

ularly its vast Persianate subset, where occultist traditions enjoyed smoother continuity and wider practice. Along with astrologers and experts in *jafr*, geomancers (sg. *rammāl*) were in high demand at imperial and regional courts during the early modern period – witness Ḥaydar Rammāl’s successful and influential career at the court of Sultan Süleymān (r. 926–974/1520–1566) in Anatolia, that of Hidāyat Allāh Munajjim-i Shirāzī at the court of Emperor Akbar (r. 963–1014/1556–1605) in India, or that of Jalāl al-Dīn Munajjim-i Yazdī, author of the *Tārīkh-i ‘Abbāsī*, at the court of Shah ‘Abbās (r. 995–1038/1587–1629) in Iran.¹² Well into the modern period we find vigorous testimony that the science was still considered a scholarly staple from the Maghrib to India; in 13th/19th-century Samarkand and Bukhara, for instance, surviving miscellany notebooks kept by judges suggest that they often employed geomantic readings to help them decide court cases.¹³ Geomancy’s popularity remains unabated in Iran today, though frequent abuse by fraudsters has made its practice into something of a social problem; as a result, *rammālī*, ‘geomancing,’ is now legally punishable by a fine and up to seven years’ imprisonment.¹⁴ In its current, flabby Persian usage *rammālī* usually connotes witchcraft or hocus pocus, particularly of the hornswoggle variety, though is not as wholly negative as the term *jādūgari*.¹⁵ (In 2009, for instance, opposition candidate Mir Hossein Mousavi derided Mahmoud Ahmadinejad’s government as one of *kaf-bīnī u rammālī*, or palmistry and poppycock.¹⁶) Yet even such infamy is an index of the continued inextricability of geomancy from Persianate culture.

The work of Emilie Savage-Smith, Marion B. Smith and Toufic Fahd in particular marks the indispensable starting point for any inquiry into Islamicate geomancy.¹⁷ Robert Jaulin has explored the mathematical properties of the science from a

1997; Marie-Luise von Franz, *On Divination and Synchronicity: The Psychology of Meaningful Chance*, Toronto: Inner City Books, 1980. More recently, however, geomancy appears to be experiencing a modest revival in the Anglo-American world; witness John Michael Greer’s *The Art and Practice of Geomancy: Divination, Magic, and Earth Wisdom of the Renaissance* (San Francisco: Weiser 2009), a comprehensive and highly usable manual in the early modern occult-philosophical tradition.

¹² On these three geomancers see below.

¹³ They also appear to have regularly used lettrist techniques to analyze people’s character on the basis of their names. More generally, biographical dictionaries for 13th/19th-century Transoxania indicate that some five percent of scholars were known to be practicing occultists – the same percentage that are identified as physicians. My thanks to James Pickett for bringing this evidence to my attention; see now our “Mobilizing Magic: Occultism in Central Asia and the Continuity of High Persianate Culture under Russian Rule”, in: *Studia Islamica* 111/2 (2016), 1–54.

¹⁴ Alireza M. Doostdar, “Fantasies of Reason: Science, Superstition, and the Supernatural in Iran”, PhD dissertation, Harvard Univ., 2012, 48 n. 19; and see now his *The Iranian Metaphysicals: Explorations in Science, Islam, and the Uncanny*, Princeton: Princeton Univ. Press (forthcoming 2017).

¹⁵ Doostdar, “Fantasies of Reason”, 46 n. 17.

¹⁶ Doostdar, “Fantasies of Reason”, 4.

¹⁷ See above; other studies include Felix Klein-Franke, “The Geomancy of Aḥmad b. ‘Alī Zunbul: A Study of the Arabic Corpus Hermeticum”, in: *Ambix* 20/1 (March 1973), 26–35;

structuralist standpoint,¹⁸ and a number of anthropologists have conducted surveys of geomantic practice in modern-day sub-Saharan Africa and Yemen, including derivative forms such as *ifa*, *gara* or *sikidy*.¹⁹ What has been almost wholly neglected to date, however, is the *Persianate* geomantic tradition.²⁰ This vital and productive tradition was the driving force behind the florescence of Islamicate geomancy from the 8th/14th century onward, itself part of the great renaissance of occultism throughout the post-Mongol Islamicate world, and appears to begin in earnest in the mid-7th/13th century with the importation to Iran of the Maghribi geomantic tradition associated in the first place with al-Zanāti. This tradition would remain the gold standard for centuries: the 7th/13th-century Shirazi geo-

Valerio Cappozzo, "Libri dei sogni e geomanzia: la loro applicazione letteraria tra Islam, medioevo romanzo e Dante", in: *Quaderni di Studi Indo-Mediterranei* 2 (2009), 207–226.

- ¹⁸ E.g., Robert Jaulin, *La géomancie: analyse formelle*, Paris: Mouton, 1966; Robert Jaulin, *Géomancie et Islam*, Paris: Bourgois, 1991. Smith's "The Nature of Islamic Geomancy" is a critique of Jaulin's studies.
- ¹⁹ E.g., Bernard Maupoil, "Contribution à l'étude de l'origine musulmane de la géomancie dans le Bas-Dahomey", in: *Journal de la Société des Africanistes* 13 (1943), 1–94; Jacques Faublée, "Techniques divinatoires et magiques chez les Bara de Madagascar", in: *Journal de la Société des Africanistes* 21 (1951), 127–138; Robert Jaulin, "Essai d'analyse formelle d'un procédé géomantique", in: *Bulletin de l'Institut Français d'Afrique Noire*, series B, 19 (1957), 43–71; William R. Bascom, *Ifa Divination: Communication between Gods and Men in West Africa*, Bloomington: Indiana Univ. Press, 1969; Philip M. Peek, ed., *African Divination Systems: Ways of Knowing*, Bloomington: Indiana Univ. Press, 1991; Annick Regourd, "Pratiques de géomancie au Yémen", in: *Religion et pratiques de puissance*, Albert de Surgy, ed., Paris: L'Harmattan, 1997, 105–128; Jan Jansen, "Éducation arithmétique sous forme d'apprentissage: la géomancie dans les Monts Mandingues", in: *Cahiers d'Études africaines* 51/1, 201 (2011), 9–49; see Smith, "The Nature of Islamic Geomancy", 9.
- ²⁰ On the almost total neglect of Persianate occultism see Francis Richard and Živa Vesel, "Un domaine méconnu: les écrits occultes en persan", in: *Bulletin d'Études Orientales* 44 (1993), 211–215. Apart from the early 20th-century surveys of Persian folklore by Henri Massé (*Persian Beliefs and Customs*, Charles Messner, transl., New Haven: Human Relations Area Files, 1954; on *raml* see 249–250) and Bess Allen Donaldson (*The Wild Rue: A Study of Muhammadan Magic and Folklore in Iran*, London: Luzac, 1938; on *raml* see 194–195), which are largely incognizant of more elite, scholarly forms of geomantic practice, the only attention that has been paid to the Persianate geomantic tradition is C. A. Storey's list of works on the subject (*Persian Literature: A Bio-Bibliographical Survey*, vol. 2, part 3, Leiden: Brill, 1977, 480–489; but as a strictly preliminary foray it is unsystematic and drastically incomplete, and focuses primarily on Indian collections. Emilie Savage-Smith and Marion B. Smith have noted the existence of a healthy Persianate geomantic tradition from Fakhr al-Dīn Rāzī and Naṣīr al-Dīn Tūṣī onward but have dealt to date primarily with Arabic geomancy, and only to the 7th/13th century. This general neglect has largely to do with the fact that Persian geomantic texts were not transmitted to Europe, as well as the fact that they are an important component of the renaissance of occultism in the early modern Islamicate world, until recently assumed by scholars to have been a mark of terminal cultural decline – the intense occultism of the European Renaissance being an exception to this rule, of course (a classic expression of such whiggish chauvinism is Armand Abel, "La place des sciences occultes dans la décadence", in: *Classicism et déclin culturel dans l'histoire de l'Islam*, R. Brunschvig and G. E. von Grunebaum, eds., Paris: Editions Bessou, 1957, 291–311). An index of the popular vitality of the Persianate geomantic tradition is its presence in *The Thousand and One Nights*; for a list of the relevant tales see Savage-Smith, and Smith, "Islamic Geomancy", 220 n. 40.

mancer Nāṣir b. Ḥaydar cleaves to the method of the Berber master, and even three centuries later the Isfahani-Tabrizi geomancer Ḥaydar b. Muḥammad felt it necessary to mention that his brilliant teacher in the science was from the Maḡhrib.²¹ More famously, Naṣir al-Din Ṭūsī (d. 672/1274), the indefatigable philosopher, mathematician and astronomer-astrologer and founder of the Maragha Observatory, wrote one of his two works on geomancy in both Arabic and Persian versions at the request of his Mongol patron Hülegü (r. 654–663/1256–1265), and is frequently cited as an authority in later Persian geomantic works.²² With Ṭūsī as precedent, geomancy went on to exercise some of the best minds of the early modern Persianate world as a mainstream occult-scientific tradition.

When did this science first strike root in the persophone world? How did it acquire such prestige and maintain it over so many centuries? Who were its major exponents? And why has its Persianate iteration been so resolutely ignored in history of science? Such are the questions this article endeavors to answer. This being the first study of the specifically Persianate geomantic tradition, however, I will limit myself to providing a preliminary, tentative survey of that tradition during its heyday between the 7th–10th/13th–16th centuries; I will also attempt to identify the philosophical-scientific reasons for its perennial intellectual appeal and continued vitality down to the present. Before embarking on these tasks, however, it will first be helpful to briefly define geomancy as a practice.

Islamicate geomancy

The various Arabic terms for geomancy, the ‘science of sand’ (*‘ilm al-raml*, *khatt al-raml*, *ḍarb*, *ṭarq*),²³ all refer to its basic procedure of drawing 16 random series of lines in the sand or dirt to generate the first four tetragrams of a geomantic reading. (Confusingly, in modern English usage geomancy can also refer to the Chinese art of *feng shui*, though this is a misnomer; as a system of divining the subtle currents of the earth for the purposes of building or burying, it is more accurately termed ‘topomancy.’²⁴) As with *I Ching* trigrams, the four lines of a geomantic figure (*shakl*) are generated by the odd (*fard*) or even (*zawj*) result of each line, creating a binary code represented as either one dot (*nuqta*) or two dots

²¹ MS Majlis 12534, p. 3. On both geomancers see below. Cf. the Persian translation by one Muḥammad b. Maṣṣūr, made in 673/1275, of an Arabic geomantic text titled *Mafātiḥ al-Asrār*; therein the translator states that he has been sojourning in Anatolia and the Maḡhrib and among the Berbers (MS Mar‘ashi 2114).

²² Emilie Savage-Smith, “Geomancy in the Islamic World”, in: *Encyclopaedia of the History of Science, Technology, and Medicine in Non-Western Cultures*, Helaine Selin, ed., Berlin: Springer, 2008, 998–999; Savage-Smith and Smith, “Islamic Geomancy”, 216; and see below.

²³ Toufic Fahd, “Khatt”, in: *Encyclopaedia of Islam*, 2nd edition, vol. 4, Leiden: Brill, 1991, 1128–1130. *‘Ilm al-raml* is occasionally translated as *‘ilm-i rig* in Persian.

²⁴ Skinner, *Geomancy*, 36. By analogy with *feng shui*, exponents of ley-line theory also adopted the term geomancy for their practice, further muddying the waters.

respectively – hence the science’s alternative name of *‘ilm al-nuqta* or *‘ilm al-niqāt*.²⁵ This binary code is then deployed according to set procedures to capture the flux patterns of the four elemental energies (fire, air, water, earth) as a means to divine past, present and future events. Emilie Savage-Smith summarizes the geomantic method as follows:

The divination is accomplished by forming and then interpreting a design, called a geomantic tableau, consisting of 16 positions, each of which is occupied by a geomantic figure. The figures occupying the first four positions are determined by marking 16 horizontal lines of dots on a piece of paper or a dust board. Each row of dots is examined to determine if it is odd or even and is then represented by one or two dots accordingly. Each figure is then formed of a vertical column of four marks, each of which is either one or two dots. The first four figures, generated by lines made while the questioner concentrates upon the question, are placed side by side in a row from right to left. From these four figures the remaining 12 positions in the tableau are produced according to set procedures. Various interpretative methods are advocated by geomancers for reading the tableau, often depending upon the nature of the question asked.²⁶



Figure 1: A typical geomantic tableau²⁷

From right to left, the first four figures in the top row are termed Mothers (*um-mahāt*), which are combined to produce the second four in the same row, termed Daughters (*banāt*); the four figures the Mothers and Daughters produce in the next row are termed Nieces (*ḥafidāt*, *mutawallidāt*); and the Nieces are combined to produce the *zawā'id* in the rows below: first, the Right and Left Witnesses (*sharwāhid*), which in turn produce the Judge (*mizān*) in the 15th position at the bottom of the geomantic tableau or shield chart (*takht*). In the 16th and final position, usually drawn below and to the right of the Judge or otherwise bracketed off, is the Reconciler (*‘āqiba*), produced by the combination of the first Mother and the Judge.

²⁵ As Stephen Skinner notes in his manual on the subject (*Geomancy*, 35): “In this century when computers now make many of man’s economic, political and commercial forecasts, it is easy to forget that these machines work on the same principle of binary mathematics as the infinitely more ancient machines of the *I Ching* and geomancy.”

²⁶ Savage-Smith, “Geomancy in the Islamic World”, 998.

²⁷ This example is from the margin of Princeton’s copy of Shams al-Din Khafri’s *R. dar Raml*, MS New Series 1177/2, f. 24b. Note that, as here, a line of two dots in a geomantic figure is usually written as a dash.

The number of possible combinations of figures in a geomantic tableau is 16⁴, or 65,536 in all.²⁸ Each of the 16 geomantic figures and the 15 houses of a tableau acquired specific elemental, astrological, calendrical, numerical, lettrist, humoral, physiognomical and other correspondences; detailed information can thus be derived from the figures about virtually any aspect of human experience, whether physical, mental or spiritual, whether past, present or future. And like the data of human experience, a figure's significance can vary depending on the context of the larger reading and the nature of the question asked: negative, constrictive figures such as *'uqla*, or Carcer, can be quite favorable for situations in which binding or limitation is desired, such as in striking a contract, and positive, expansive figures like *ahyān*, or Laetitia, unfavorable for situations requiring stability or the keeping of secrets. The 16 geomantic figures are as follows (with their standard Latin names in square brackets):

⋮	Way (<i>tariq</i>) ²⁹ [Via]	⋮⋮	Group (<i>jamā'a</i>) ³⁰ [Populus]
⋮⋮	Union (<i>ijtimā'</i>) ³¹	⋮⋮	Binding (<i>'uqla</i>) ³² [Carcer]
⋮⋮	[Conjunctio]	⋮⋮	
⋮⋮	Internal help (<i>nuṣra dākhiba</i>) ³³	⋮⋮	External help (<i>nuṣra khārija</i>)
⋮⋮	[Fortuna Major]	⋮⋮	[Fortuna Minor]
⋮⋮	Internal seizing (<i>qabḍ dākhib</i>) ³⁴ [Acquisitio]	⋮⋮	External seizing (<i>qabḍ khārij</i>)
⋮⋮		⋮⋮	[Amisio]
⋮⋮	Propitious times (<i>ahyān</i>) ³⁵	⋮⋮	Inversion (<i>inkis</i>) ³⁶ [Tristitia]
⋮⋮	[Laetitia]	⋮⋮	
⋮⋮	White (<i>bayāḍ</i>) ³⁷ [Albus]	⋮⋮	Red (<i>ḥumra</i>) ³⁸ [Rubeus]

²⁸ Smith, "The Nature of Islamic Geomancy", 14.

²⁹ Hidāyat Allāh Munajjim-i Shirāzi, on whom more below, translates this as *rāb raftan*, 'going' (*Qavā'id al-Hidāya*, MS Majlis 12563, p. 339).

³⁰ Hidāyat Allāh translates this as *khalq-i bisyār u qaṣr-i bisyār*, 'many people, great restraint.'

³¹ Hidāyat Allāh translates this as *payvastagi bi bar chi maṭlūb bāshad*, 'attainment of whatever is desired.'

³² Alt. *thikāf*, 'straightening instrument, swordfighting'; *shiqāf*, 'clay pots'; *ītikāf*, 'withdrawal, seclusion.' Hidāyat Allāh translates this as *band u bastagi*, 'bond and binding.'

³³ Or 'increasing victory,' contrasted with *nuṣra khārija* as 'diminishing victory.' Alt. *tashmir*, 'readying'; *al-sultān*, 'the ruler.' Hidāyat Allāh translates the first as *yāri dabanda dar āmadan*, 'giving help in coming,' and the second as *yāri dabanda dar kburūj*, 'giving help in going.'

³⁴ Or 'internal prize,' contrasted with *qabḍ khārij* as 'external prize.' Hidāyat Allāh translates the first as *bi dast āranda*, 'bringing into hand,' and the second as *az dast baranda*, 'taking from hand.'

³⁵ Alt. *lahyān*, 'jaw'; *hiyān*, ? (translated by Ibn Maḥfūf as *kathir al-bayba* and *al-rajul al-kabir*); *ḍāḥik*, 'laughing.' Hidāyat Allāh translates *lahyān* as *riṣh-i darāz*, 'longbeard.'

³⁶ Alt. *mankūs*, 'inverted, overturned, relapsed'; *nakis*, 'inverting'; *rakiza*, 'pole, peg, buried treasure.' Hidāyat Allāh translates this as *sar-nigūn shudan*, 'becoming inverted, depressed.'

³⁷ Alt. *laban*, 'milk.' Hidāyat Allāh translates this as *chiz-bā-yi safid*, 'white things.'

³⁸ Alt. *dam*, 'blood,' or *damm*, 'rouge.' Hidāyat Allāh translates this as *chiz-bā-yi surkh*, 'red things.'

∴	Pure-cheeked [girl] (<i>naqī l-khadd</i>) ³⁹ [Puella]	∴	[Thin-bearded boy] (<i>javādala</i>) ⁴⁰ [Puer]
∴	Inner lintel (<i>‘ataba dākhlila</i>) ⁴¹ [Caput Draconis]	∴	Outer threshold (<i>‘ataba kbārīja</i>) ⁴² [Cauda Draconis]

Table 1: The 16 geomantic figures⁴³

Although many techniques of divination long persisted in the Islamicate world,⁴⁴ *raml*, with *jafṛ* and quranic bibliomancy (*fāl-i Qurʾān*),⁴⁵ has the distinction of being the most fully Islamicized; this trio soon outpaced the others in prestige and elite patronage as a result.⁴⁶ Geomantic treatises frequently begin by marshaling a number of prooftexts from the Quran and Hadith in defense of the science’s religious legitimacy. The two most cited of these:

Bring me a Book before this, or some trace of a science, if you speak truly (Q 46:4).

There was a prophet [who practiced divination by] drawing lines [in the sand], so one may draw the way he did.⁴⁷

In the case of the first, this ‘trace of a science’ (*athārat^{an} min ‘ilm*), here suggested to be on a par with scriptural revelation, was readily linked to the geomantic

³⁹ Alt. *ashqar*, ‘fair’; *awwāḥ*, ‘modest ones.’ Hidāyat Allāh translates this as *pāk-rūy u sāda*, ‘smooth-faced and beardless, pure.’

⁴⁰ Alt. *karwasaj/kūsaj*, ‘thin-bearded.’ Hidāyat Allāh re-identifies (misidentifies?) this figure with *farah*, aka *‘ataba dākhlila*, which he translates as *shādī u khurramī*, ‘joy and gladness.’

⁴¹ Alt. *rāyat al-farah*, ‘banner of joy.’ Hidāyat Allāh translates this as *dākhlil shudan-i astāna*, ‘entering the threshold.’

⁴² Alt. *rakiza thābita*, ‘fixed pole or peg, fixed treasure.’ Hidāyat Allāh translates this as *birūn raftan az astāna*, ‘leaving the threshold.’

⁴³ For a discussion of these figures and their various names see Savage-Smith and Smith, “Islamic Geomancy”, 233–235.

⁴⁴ Scapulomancy (*‘ilm al-katīf*), for instance, or divination by shoulder blade, was frequently used in Khorezm until the modern period to predict the outcome of caravan journeys, resolve legal cases, etc. (personal communication with James Pickett).

⁴⁵ Bibliomancy, especially in reference to the Quran or the *Divān* of Hāfiz (d. ca. 792/1390), similarly flourished in the Persianate world during the 7th–10th/13th–16th centuries; see e.g. Christiane Gruber, “The ‘Restored’ Shi‘i *muṣḥaf* as Divine Guide? The Practice of *fāl-i Qurʾān* in the Safavid Period”, in: *Journal of Qurʾanic Studies* 13/2 (2011), 29–55.

⁴⁶ Fahd, *La divination arabe*, 204.

⁴⁷ *Kāna nabīyy^{um} minā l-anbiyāʾi yakhuṭṭu fa-man wāfaqa khaṭṭa-hu fa-dhāk*. See e.g. Muslim, *Ṣaḥīḥ*, *Kitāb al-Masājīd wa-Mawāqīʿ al-Ṣalāt*, bab 8, no. 1227; Abū Dāwūd, *Sunan*, *Kitāb al-Ṣalāt*, bab 173, no. 931; etc. Some hadith commentators acknowledge this as a prooftext for the licitness of geomantic practice, but a number, including al-Nawawī, al-Khaṭṭābī and al-Qāḍī ʿIyāḍ, argue for its illicitness during the Islamic dispensation; see e.g. Muḥammad al-Amin al-Urmī, *al-Kawākib al-Wahbāʾ wa-l-Rawḍ al-Bahbāʾ fī Sharḥ Ṣaḥīḥ Muslim b. al-Ḥajjāj*, 26 vols., Jeddah: Dār al-Minhāj, 2009, vol. 8, 126–128. Ibn Rushd al-Jadd (d. 520/1126) likewise judges geomancy to be illicit in his fatwa on the subject, attempting to demonstrate that this hadith in fact represents not an endorsement but a definitive prohibition (*Fatāwā Ibn Rushd*, al-Mukhtār b. al-Ṭāhir al-Talili, ed., 13 vols., Beirut: Dār al-Gharb al-Islāmī, 1407/1987, vol. 1, 249–261).

procedure of tracing lines in the sand.⁴⁸ And in the second, an authentic (*ṣaḥīḥ*) hadith, the prophet in question was usually taken to be Daniel (or Idris). A third hadith ascribed to ʿAlī b. Abī Ṭālib, fountainhead of sacral power (*walāya*) and patron saint of Islamicate occultism, was likewise popular (*ṭawīl* meter):

Reck you not that God (His praise be exalted)
effluxed upon Enoch a sandy science (*ʿilm^{an} murammal^{an}*)
and through His grace inspired him with the science of dots (*ʿilm al-niqāt*),
revealing to him what had ere been obscure?⁴⁹

Persianate geomancy

Perhaps the best overview of the premodern and modern Islamicate as well as the specifically Persianate geomantic tradition is that given by Muḥammad Bāqir Ṭabāṭabāʾī Yazdī in his *Nafahāt al-Asrār*, written in Damascus in 1277/1861.⁵⁰ This work, while primarily in Arabic, favors Persian sources, though without neglecting the standard Arabic texts on the subject. In the course of two prefaces, ten chapters and two epilogues, Muḥammad Bāqir provides a comprehensive introduction to the science, liberally referencing or quoting his sources; most usefully for our purposes here, the second epilogue is given over to a list of the 54 texts or authors that he drew upon when writing his own book: the geomantic canon. It should be noted, however, that his list is copied almost verbatim from the list of 50 titles given by Hidāyat Allāh Munajjim in his *Qavāʿid al-Hidāya*, adding only three additional works to the end (*Asrār al-Mafātīḥ*, *Irsbād al-Raml* and the *Qavāʿid al-Hidāya* itself); this suggests a lack of new developments in the science between the 11th/17th and 13th/19th centuries.⁵¹ (The list of 49 geomantic

⁴⁸ On this verse see e.g. Muḥammad al-Qurṭubī (d. 671/1272), *al-Jāmiʿ li-Aḥkām al-Qurʾān*, 20 vols., Tehran: Nāṣir-i Khusraw, 1364 Sh./1985, vol. 16, 179–183; the exegete here argues for the licitness of oneiromancy and omen interpretation or bibliomancy (*ṣāʾl*), but the illicitness of other forms of divination, including ornithomancy (*tīra*) and augury (*zajr*) (and presumably geomancy).

⁴⁹ *A-lam tara anna Llāha jalla thanā-hu / afāda ʿalā Idrisa ʿilm^{an} murammal^{an} // wa-alhama-hu ʿilma l-niqāṭi bi-faḍli-hi / wa-awḥā ilay-hi mā kāna mushkī^{an}.*

⁵⁰ His full name is Muḥammad Bāqir b. Murtaḍā Ḥasani Ḥusayni Ṭabāṭabāʾī Yazdī (also Ḥāʾiri), resident of Najaf. The *Nafahāt al-Asrār* was published in lithograph in Najaf in 1359/1940, my source here.

⁵¹ Ṭabāṭabāʾī Yazdī, or at least the Najaf lithograph of the *Nafahāt*, also introduces a few errors into Hidāyat Allāh's list: it mistakenly splits the *Abwāb-i Anwār* into two works, has *Uṣūl* for *Ḥuṣūl* in two titles, drops one title (*risāla-yi Kulab-i Kabūd*), moves the *Jāmiʿ al-Asrār* from its place at the end, and most importantly does not seem to be aware that *al-Wāfi* is the title of Ṭūṣī's treatise. The full *Nafahāt* list is as follows, with corrections in brackets (116): *Miftāḥ al-Kunūz*, *Kanz al-Daqaʿiq*, *al-Shajara wa-l-Thamara*, *Shāmil al-Uṣūl* [*al-Ḥuṣūl*], *al-Tuḥfa al-Shābiyya*, *Uṣūl al-Raml*, *ʿAyn al-Raml*, *Shamʿ al-Raml*, *Miṣbāḥ* [*Miftāḥ*] *al-Mafātīḥ*, *al-Wāfi*, *al-Si Bāb*, *al-Fuṣūl*, *al-Abwāb*, *al-Anwār* [*Abwāb al-Anwār*], [the work of] al-Uqlidi, [the work of] al-Ṭarābulṣi, [the work of] al-Rawshani, [the treatise of Kulah-i Kabūd], *Kashf al-Asrār*, *Jāmiʿ al-Asrār*, *al-Talkhīṣ*, the treatise of Abū l-ʿAbbās al-Būni,

works given by Ḥājji Khalifa does not appear to be a source.⁵²) Space does not permit a full discussion of these here, and due to the extreme paucity of research on the subject I have been unable to identify the authors of many of the listed titles in any case; I will accordingly limit myself to listing, in chronological order, a number of what appear to be representative authors and works as a basis for further research:

1. Abū ‘Abd Allāh Muḥammad al-Zanāti (fl. before 629/1230), *al-Fuṣūl* and *Shajarat al-Raml*

This Berber scholar, again, would seem to be the most cited of early authorities on geomancy, even in the Persianate east. The first work, also titled *al-Faṣl fī Uṣūl al-Raml*, has remained popular down to the present, as testified to by a number of undated Cairo bazaar editions; one produced by Maktabat al-Qāhira, for example, is paired with a treatise on *jafr* attributed to Ja‘far al-Šādiq. A Persian translation was done into Greek by the monk Arsenius in 1266;⁵³ a Turkish translation is entitled *Hulūl al-Asbkāl*.

For its part, the *Shajara* appears to have inspired a long tradition of Persian commentary and supercommentary under the rubric *Shajara u Thamara*. Works by the latter title are also said to be based on a Greek original, attributed to Daniel, which was translated into Arabic by Māshā’allāh al-Baṣrī and subsequently into Persian; a Turkish translation, entitled *al-Durr al-Manthūr*, was similarly made.⁵⁴ Another Persian commentary, entitled *Kashf al-Shajara*, was produced in 913/1507 by one Maḥmūd Kawtharī b. Ismā‘īl for Muḥammad Shībānī Khan (r. 906–918/1500–1512).⁵⁵

Minbāj al-Asrār, *Jāmi‘ al-Kifāya*, *al-Kbulāša*, *Kbulāṣat al-Baḥrayn*, *al-Tabdīb*, *al-Dhakhira*, *al-Zubda*, *Natijat al-‘Ulūm*, *al-Qawā‘id*, *Nuzhat al-‘Uqūl*, *al-Jāmi‘*, *al-Kāmil of al-Ustād Ḥusayn al-Qaffāl*, *Riyād al-Tālibin*, *Shajarat al-Awraq*, *Sā‘ir al-Awliyā’*, *al-Miṣbāḥ*, *al-Miftāḥ*, *Hidāyat al-Nuqta*, *R. Surkhāb*, [the work of] al-Biyābānaki, [the work of] al-Bādkūhi [Bābā Kūhi], *al-Tawdīḥ*, *al-Takwīḥ*, the work of Mullā Shāh, *Taqwīm al-Raml*, *Aṣl al-Mafātīḥ*, *Zayn al-Raml*, *Kāmil al-Uṣūl [al-Huṣūl]*, *Jabān al-Raml*, *Asrār al-Mafātīḥ*, *Irshād al-Raml*, *Qawā‘id al-Hidāya* and the treatise of Fakhr al-Muḥaqqiqin al-Khwāja Naṣir al-Dīn [= *al-Wāfi*].

⁵² While it accords in the main with those by Hidāyat Allāh and Tabāṭabā‘ī Yazdī, Ḥājji Khalifa’s list is more problematic; it mistakenly alters (*Mafātīḥ al-Kumūz* for *Miftāḥ al-Kumūz*, *Kifāya* for *Jāmi‘ al-Kifāya*, *Shajara-yi Awzān* for *Shajara-yi Awraq*, *Rawshan* for *Rawshani*), recombines (*Alwāb al-Raml* and the *Anwār of Uqlidi* for *Alwāb al-Anwār* and an untitled work by Uqlidi) or duplicates (two *Miṣbāḥs*) certain titles, while also adding five new works: the work of Mawlānā Pashsha, *R. Yūnus*, the *Kumūz* of Abū ‘Alī, *Lubāb al-Lubāb* and Ibn Maḥfūf’s *Raml al-Muqawwim*. See *Kashf al-Zunūn ‘an Asāmi l-Kutub wa-l-Funūn*, 2 vols., vol. 1, Beirut: Dār Iḥyā’ al-Turāth al-‘Arabi, [n.d.], cols. 912–913.

⁵³ Skinner, *Geomancy*, 63.

⁵⁴ Āqā Buzurg Tīhrānī, *al-Dharī‘a ilā Taṣānīf al-Shī‘a*, Beirut: Dār al-Aḍwā’ 1983, vol. 13, 41, no. 131. Note that Māshā’allāh is not identified as a translator from Greek in the early sources (my thanks to Sonja Brentjes for this observation).

⁵⁵ Storey, *Persian Literature*, vol. 2.3, 480, no. 848; Āqā Buzurg, *al-Dharī‘a*, vol. 18, 40; see MS Malik 5868 and MS Asiatic Society of Bengal 1516.

2. Abū l-‘Abbās al-Būnī (d. 622/1225?), *risāla*

The provenance of this untitled treatise is uncertain at best; I am not aware of a geomantic work ascribable to al-Būnī. Even the *Shams al-Ma‘ārif al-Kubrā*, the seminal and otherwise comprehensive 10th/16th-century grimoire loosely based on al-Būnī’s authentic works, does not appear to contain any material on the subject.⁵⁶ Nevertheless, that al-Būnī was increasingly cited as an authority on geomancy is testament to the confluence of that science with lettrism in the early modern Persianate world, as will be shown below.

3. Naṣīr al-Dīn Ṭūsī (d. 672/1274), *al-Vāfi fi ‘Ilm al-Raml*

This short treatise, dedicated by the author to Hülegü – whose powers of analysis are here described as ‘alchemically potent’ (*kīmiyā-athar*) –, exists in both Persian and Arabic versions, the first typically titled *al-Vāfi fi ‘Ilm al-Raml* and the second *al-Sulṭāniyya fi l-Raml*. Both versions are well represented in the manuscript record, particularly the Persian (preserved in at least 20 MS copies in Iran alone), and there is little reason to doubt the authenticity of the treatise’s ascription to Ṭūsī.⁵⁷ A facsimile copy was printed in *Majmū‘a-yi ‘Ilm al-Raml* (Qom/Urumiye, n.d.). The copy in MS Dānishgāh 6315 is followed by a brief treatise on the population of *budūh dā‘iras*. A lengthy commentary on the Persian version was written by Nūr al-Dīn Faṭḥ Allāh Abhari (fl. 786/1384), titled simply *Sharḥ-i Risāla-yi Raml-i Khwāja Naṣīr*.⁵⁸ The same commentator states that Ṭūsī originally wrote his treatise in Turkic, after which another scholar translated it into Persian (a rather dubious claim). It was versified in 169 lines by one ‘Azizī (Mashhad, MS ‘Abd al-Majid Mawlavī 446).

Another Persian treatise on geomancy is attributed to Ṭūsī under the title *Aḥkām-i Davāzdah Khāna-yi Raml*.⁵⁹ Likewise, an *Asrār-i Mafātiḥ dar Raml* (Prince-

⁵⁶ Savage-Smith and Smith, “Islamic Geomancy”, 39. On the Būnian corpus see Noah Gardiner, “Esotericism in a Manuscript Culture: Aḥmad al-Būnī and His Readers through the Mamlūk Period”, PhD dissertation, Univ. of Michigan, 2014.

⁵⁷ See Muḥammad-Taḳī Mudarris Raḍavi, *Aḥwāl u Āthār-i ... Muḥammad b. Muḥammad b. al-Ḥasan al-Ṭūsī Mulaqqab bi Khwāja Naṣīr al-Dīn*, Tehran: Univ. of Tehran Press, 1334 Sh./1955), 295–297: 309 (the Persian and Arabic versions of *al-Vāfi* are here mistakenly treated as different works); Storey, *Persian Literature*, vol. 2, 3, 480, no. 845; Āqā Buzurg, *al-Dharī‘a*, vol. 11, 250, no. 1535; Carl Brockelmann, *Geschichte der arabischen Litteratur*, 3 vols. and 2 supplemental vols., Leiden: Brill, 1937–1942, vol. 1, 676 no. 56; suppl. vol. 1, 933. It should be noted, however, that the Ilkhanid *Safīna-yi Tabriz*, which contains some material on geomancy and various works by Ṭūsī, does not include a copy in its present incomplete state (Živa Vesel, “La représentativité des matériaux scientifiques dans Safīna-ye Tabriz: le cas de minéralogie et des sciences occultes”, in: *The Treasury of Tabriz: The Great Il-Khanid Compendium*, A. A. Seyed-Gohrab and S. McGlinn, eds., Amsterdam: Rozenberg, 2007, 207–220: 213 n. 306).

⁵⁸ Āqā Buzurg, *al-Dharī‘a*, vol. 8, 234, no. 977; see e.g. MS Majlis 12509/2, ff. 19a–139b.

⁵⁹ Mudarris Raḍavi, *Aḥwāl*, 296.

ton, MS New Series 427) is misattributed to Ṭūsī under the title *Kitāb-i Shamar-i Shajar* (i.e., the *Shajara u Thamara* discussed above in connection with al-Zanāti); more than one later *Asrār al-Mafātīḥ* is extant. In one copy made for Mirzā Muḥammad Karīm Khān Nūrī (MS Marʿashī 8979/1), a note attributes this work to Ṭūsī, while the *Fibrīst-i Mushtarak-i Pākistān* attributes it to ʿAlī b. Khalīl al-Ṭarābulusī, and the Marʿashī catalog attributes the base text *Shajara* (MS Marʿashī 4229/2) to one Mullā Muḥammad Ḥusayn Hamadāni.

4. Nāṣir [al-Dīn b.] Muḥammad b. Ḥaydar Rammāl-i Shirāzī (7th/13th c.),
Maḥmūd al-Raml

Some confusion exists as to this geomancer's name, which variously appears as Nāṣir al-Dīn Muḥammad or Nāṣir b. Muḥammad,⁶⁰ as well as to the title of his most popular work, of which at least nine MSS survive in Iran. *Maḥmūd al-Raml* seems to be used most frequently, but alternative titles include *Nuzbat al-ʿUqūl va Nihāyat al-Uṣūl*, *Irshād al-Raml*, *Tuhfat al-Raml*, *Lubāb al-Raml* and *Nāṣir al-Raml*, or simply *R. Nāsiri*; the *Irshād al-Raml* cited by Ṭabāṭabāʾī Yazdī is almost certainly the same work. This treatise is presumably an adaptation of al-Zanāti's *K. al-Faṣl* in particular (in the introduction the author assures us that he in no way departs from the method of the Berber master),⁶¹ and was written for one Sayyid Ḥusayn b. ʿAlī ʿAlavī. Nāṣir is also credited with the astrological work *Hidāyat al-Nujūm*, and an *Irshād* on the astrolabe (perhaps a source for the above confusion of titles). Another popular geomantic manual, *Tuhfa-yi Shābi*, was penned by one Aḥmad b. Ḥaydar b. Muḥammad Shirāzī⁶² – perhaps Nāṣir's brother or relative, or even Nāṣir himself. (The currency of this work is testified to by the fact that a copy is preserved in the deluxe geomantic *majmūʿa* MS Majlis 12563/2, on which more below.) Finally, it bears noting in this connection that another Shirazī Rammāl, Saʿd b. Muḥammad, was author of the geomantic work *Maṭālib al-ʿArīfin* (MS Marʿashī 1081/4).

The example of Nāṣir, in short, signals the emergence of Shiraz, and Fars more generally, as capital of the emergent Persianate geomantic tradition. It appears to have retained this status until at least the 11th/17th century, moreover; a full half of the geomantic works identified in this list were likely produced in the region or, as with *Hidāyat Allāh*, by Shirazis abroad.

⁶⁰ See Storey, *Persian Literature*, vol. 2.3, 485, no. 856 (17, 21), 486 (33), 487 (38, 39), 488 (52).

⁶¹ See e.g. MS Ganj Bakhsh 1686, p. 3.

⁶² Storey, *Persian Literature*, vol. 2.3, 485, no. 856 (21 n. 2).

5. Muʿin al-Dīn Muḥammad b. Naṣr Allāh Khāqānī, aka Mullā Muʿin (fl. 706/1307), *Zubdat al-Raml*

This modest versified treatise on geomancy is typically identified as the *zubda* of Muʿin.⁶³ Another *Zubdat al-Raml* is ascribed to Mawlavī Sayyid Rawshan ʿAlī Rammāl-i Fayḍābādī, which was printed in lithograph form multiple times: Delhi, 1292/1875 and 1293/1876; India, 1881; Lucknow, 1335/1917, as part of a collection of geomantic texts under the abovementioned name of Nāṣir b. Muḥammad Rammāl. This Rawshan ʿAlī is credited with another printed geomantic work, *Sirāj al-Raml* (Lucknow, 1876; Cawnpore, 1891 and 1908). A *Mujarrabāt dar Raml* is also attributed to a Muʿin al-Dīn, presumably the same geomancer (MS Majlis 1181/3).

6. Ḥasan b. ʿUmar Ṭabasī (fl. 735/1335), *Miftāḥ al-Kunūz fī l-Rumūz*

The *Miftāḥ al-Kunūz* represents a milestone in the development of a specifically Persianate geomantic tradition: Hidāyat Allāh reports that in it Ṭabasī establishes an entirely new method of geomancy based on what he called the Most Complete (*aṣaḥḥ*) cycle, wholly ignoring the three well-known methods of his day, including that associated with al-Zanātī.⁶⁴ It is preserved in at least 13 MS copies and partial copies in Iran, and in the deluxe *majmūʿa* MS Majlis 12563/5 immediately follows the *Qavāʿid al-Hidāya*. As Hidāyat Allāh also reports, Ṭabasī's teacher in geomancy was one Qāḍī ʿAbd al-Raḥmān Jinnī.⁶⁵ His *Aḥkām*, another work on geomancy, was commented on by one Muḥammad Munajjim-i Iṣfahānī.⁶⁶

Āqā Buzurg reports that Ṭabasī wrote the *Miftāḥ al-Kunūz* in Yazd in 735/1335, and in 968/1561 the Safavid shah Ṭahmāsb (r. 930–984/1524–1576) requested of Muḥammad b. Jalāl (or Jamāl) b. ʿAbd Allāh al-Ḥusaynī (or al-Ḥasani) al-Balyānī, a geomancer connected to his court (to be identified with ʿAbd Allāh Balyānī below), that he send him a work on geomancy; Sayyid Balyānī accordingly sent him a copy of the *Miftāḥ al-Kunūz* with a new introduction, as the original was missing.⁶⁷ (Muḥammad-Taqī Dānishpazhūh notes that the *Miftāḥ al-Rumūz wa-Miqlād al-Kunūz* of one Ḥasan b. Zāhid Gharīb-i Kirmānī, written in 726/1326 for the Delhi sultan Muḥammad Shāh II b. Tughluq (r. 725–752/1325–1351), may well be the same work.⁶⁸) Āqā Buzurg also flags the presence of another geomancer at

⁶³ Storey, *Persian Literature*, vol. 2.3, 480, no. 846, and 489, no. 856 (58); cf. Āqā Buzurg, *al-Dharīʿa*, vol. 19, 195, no. 878 (he notes that a work by this title printed in Delhi is attributed to Ibn Sinā).

⁶⁴ *Qavāʿid al-Hidāya*, MS Majlis 12563/4, p. 316.

⁶⁵ *Qavāʿid al-Hidāya*, MS Majlis 12563/4, p. 323.

⁶⁶ See e.g. MS Majlis 12509/3, f. 139b.

⁶⁷ *Al-Dharīʿa*, vol. 21, 344–345, no. 5392. Three MS copies of the *Miftāḥ al-Kunūz* appear to preserve Sayyid Balyānī's version.

⁶⁸ Muḥammad-Taqī Dānishpazhūh, “Yak Parda az Zindagāni-yi Shāh Ṭahmāsb-i Ṣafavi”, in: *Justār-bā-yi Adabi* 28 (1350 Sh./1971), 915–997, here: 993.

Ṭahmāsb's court, one Ramzī Iṣfahānī, though he is not known to have produced a manual on the subject.⁶⁹

7. Anonymous (8th/14th c.?), *R. Surkhāb*

This treatise appears to be associated with the title *Sayr-i Nuqta*, the term for a standard geomantic procedure.⁷⁰ In geomantic lore *Surkhāb* was the name of the shepherd who became Daniel's first client (see below), and so is associated with the Danielic geomantic tradition in particular.⁷¹ The popularity of this work is confirmed by the fact that it is quoted by Ḥusayn Vā'iz Kāshifī (d. 910/1505), the great Timurid preacher, polymath and occultist, in his own seminal manual of magic, the *Asrār-i Qāsimī*, which in turn assured its currency down to at least the 13th/19th century; the fine copy preserved in the deluxe geomantic *majmū'a* MS Majlis 12563/3, for instance, is dated 1312/1895. It also survives in at least 26 other MS copies in Iran alone, and two at Princeton; a lithograph was thrice printed as *Kullīyyāt-i Kitāb-i Mustatāb-i Surkhāb-i Raml-i Mansūb bi Ḥaḍrat-i Dāni-yāl-i Payghambar* (Bombay, 1309/1892 and 1312/1894; Tehran: Shirkat-i Kānūn-i Kitāb, n.d.).⁷² Significantly for our purposes in this survey, Sayyid Muḥsin al-Amin considers *R. Surkhāb* to be the title of a geomantic work by Sayyid Ḥusayn Akhlāṭī, the great Kurdish alchemist-lettrist, on whom see below.⁷³ (The same *majmū'a*, MS Majlis 12563, notably concludes with a long, untitled work by Akhlāṭī on alchemy, further suggesting his connection, at least in its later reception, to the *R. Surkhāb*.)

8. Anonymous, *Miṣbāḥ al-Raml*

This anonymous but oft-cited Persian work in 16 chapters is sometimes associated with one Abū l-Barakāt – presumably Abū l-Barakāt al-Baghdādī (d. after 560/1164), the famed Jewish philosopher identified, probably incorrectly, as Fakhr al-Din Rāzī's teacher.⁷⁴

9. Anonymous, *Sā'ir al-Awliyā'*

This work emphasizes geomancy's status as a natural science, and is cited as a source by Shams al-Din Khafri (d. 942/1535) in particular. (Tellingly, in the Princeton copy, MS New Series 1177/1, ff. 1a–15b, it is immediately followed by

⁶⁹ *Al-Dharī'a*, vol. 9.2, 384, no. 2249.

⁷⁰ Āqā Buzurg, *al-Dharī'a*, vol. 12, 279, no. 1866.

⁷¹ It may be noted in passing that the Kurdish 'Annāzids (4th–6th/10th–12th c.) feature three dynasts with the name *Surkhāb*, and the Kāvūsi Bāwandids (1st–5th/7th–11th c.) two.

⁷² Storey, *Persian Literature*, vol. 2.3, 488, no. 856 (47).

⁷³ *A'yān al-Sbī'a*, Ḥasan al-Amin, ed., 11 vols., Beirut: Dār al-Ta'āruf, 1983, vol. 6, 94.

⁷⁴ See Ṭabāṭabā'i Yazdi, *Nafahāt al-Asrār*, 5, 10, 29, 30, 41, 100. I was unable to examine any manuscript copies of this work.

Khafri's own *R. dar Raml*, on which see below.) Hidāyat Allāh identifies it as epitomizing the eastern, i.e., Khurasani-Indian, geomantic school.⁷⁵ Its popularity is suggested by the survival of at least 11 MSS in Iran alone.

10. Zayn al-ʿĀbidin b. Qāsim Rammāl-i Iṣfahānī (8th/14th c.), *Zayn al-Raml*

This geomancer is not to be confused with Zayn al-ʿĀbidin Vāhid al-ʿAyn Rammāl-i Iṣfahānī, whose son Ṣafyā-yi Rammāl carried out a famous poetic insult match (*mubājāt*) with Shah ʿAbbās's personal physician and boon companion, Sharaf al-Dīn Ḥasan Iṣfahānī, aka Ḥakim Shifāʾi Iṣfahānī (d. 1037/1628).⁷⁶ Other titles like *Kanz al-Aḥbāb*, *Thamarāt al-Arwāḥ* and *Sharḥ-i Kitāb-i Shajara* are also associated with the former Zayn al-ʿĀbidin.⁷⁷ The *Zayn al-Raml* survives in at least three MSS in Iran.

11. ʿAbd al-Ghanī Ḥāfiẓ Shirvānī (8th/14th c.), *Anvār al-Raml*⁷⁸

A *Khulāṣat al-Raml*, attributed to the author as well, is perhaps an earlier version of the *Anvār al-Raml*. The latter treatise survives in at least 11 MS copies in Iran, as well as in one in Cairo dated 1161/1748 (MS Maʿārif al-Asrār Fārisī Ṭalʿat 2), and a number of lithograph printings exist (India, 1293/1876 and 1313/1896; Lucknow, 1335/1917; Lucknow, 1877, ed. ʿAziz al-Ḥasan; Lucknow: Nawal Kishawī, 1881 and 1896).

12. Luṭf Allāh b. ʿAbd al-Malik Nishāpūri Samarqandī (d. 816/1413?), *Khulāṣat al-Baḥrayn*

On this work and author see below.

13. Ḥasan Abarqūhī (fl. early 9th/15th c.), *urjūza*

Hidāyat Allāh, and following him Ṭabāṭabāʾī Yazdī, identifies this geomancer as the student of Sayyid Ḥusayn Akhlāṭī – a fact of some significance, as will be seen.⁷⁹ While Ḥasan Abarqūhī is not known to be the author of a geomantic manual of his own, his seven-line *urjūza* on the *istikbrāj al-ḍamīr* procedure⁸⁰

⁷⁵ *Qavāʿid al-Hidāya*, MS Majlis 12563/4, p. 364.

⁷⁶ Vālih Dāghistānī, *Tadbkira-yi Riyāḍ al-Shuʿarāʾ*, Muḥsin Nāji Naṣrābādī, ed., 5 vols., Tehran: Asāṭir, 1384 Sh./2005, vol. 2, 1195, no. 1206.

⁷⁷ Storey, *Persian Literature*, vol. 2.3, 488, no. 856 (55).

⁷⁸ Storey, *Persian Literature*, vol. 2.3, 483, no. 856 (1); Āqā Buzurg, *al-Dharīʿa*, vol. 2, 427, no. 1682.

⁷⁹ *Qavāʿid al-Hidāya*, MS Majlis 12563/4, 333; *Nafahāt al-Asrār*, 29.

⁸⁰ *Istikbrāj al-ḍamīr* is used to determine the main factor in one's question; in his discussion of geomancy, Shams al-Dīn Āmulī (on whom more below) reports that there is considerable difference of opinion on the subject and gives five different techniques, the second involving, for example, the combination of the first and the sixteenth figures of a tableau on the one hand and the fourth and seventh on the other, then the combination of the two

would seem to be well-known to later specialists. It begins, ‘Take every odd level of a reading’ (*bama-yi marātib-i aḡrād-i raml-rā bar-gīr*), and ends with an assertion of the science’s orthodoxy: ‘In this method as set forth by Ḥājji Ḥasan / nowise may the decree of the Creator be altered.’⁸¹

14. Nizām al-Dīn ‘Abd al-Qādir b. Ḥasan Rūyānī Lāhijī (d. 925/1519),
Miftāḥ-i Maḡfātīḥ

It is with such figures as ‘Abd al-Qādir Rūyānī that we begin to see the consolidation of the convergence, from Ṭūsī onward, of the professional roles of astronomer-astrologer and geomancer, such that henceforth in the Persianate world the title *munajjim* typically refers to both. This Gilani scholar was first and foremost an astronomer: as a student of ‘Alī Qūshchī (d. 878/1474), he produced a commentary, entitled *Tuhfa-yi Nizāmiyya*, on Ṭūsī’s seminal *Sī Faṣl*; the *Zij-i Mulkhkhāṣ-i Mirzā’i*, an abridged version of the landmark *Zij-i Jadid-i Sulṭānī* produced at the Timurid sultan Ulugh Beg’s (r. 811–853/1409–1449) Samarkand Observatory; and a short almanac key in the established *ma‘rifat-i taqvim* genre (as did Ṭūsī himself). While the first two works were dedicated to Kiyā’ī governors, ‘Abd al-Qādir spent part of his career teaching at the Shāh Madrasa in Herat under Sulṭān-Ḥusayn Bāyqarā (r. 873–911/1469–1506).⁸² The *Miftāḥ-i Maḡfātīḥ*, of which at least eight MSS survive in Iran, is formally a commentary on the important but brief geomantic text *Maḡfātīḥ*, whose author I have not been able to identify; while other commentaries seem to have been produced on the original treatise, with titles like *Asrār-i Maḡfātīḥ* (sometimes ascribed to Ṭūsī) and *Aṣṭ-i Maḡfātīḥ*, ‘Abd al-Qādir’s was perhaps the most popular.

That the Gilani astronomer-geomancer appears to have been a member of the Samarkand school is highly significant in the history of science context – for his teacher Qūshchī and the latter’s teacher Qāḏizāda Rūmī (d. 835/1432) were responsible for the fateful and unprecedented push to *mathematize astronomy*; they are the first in the Islamo-Christianate astronomical tradition as a whole to do so, and hence in a sense the indirect forebears of Isaac Newton himself, as has recently been shown.⁸³ This project was continued into the Safavid period,

resulting figures (*Nafāyis al-Funūn fī ‘Arāyis al-‘Uyūn*, Abū l-Ḥasan Sha‘rānī and Ibrāhīm Miyānji, eds., 3 vols., Tehran: Intishārāt-i Islāmiyya 1381 Sh./2002, vol. 3, 548–549).

⁸¹ Ṭabāṭabā’ī Yazdī, *Nafahāt al-Asrār*, 29.

⁸² Āqā Buzurg, *al-Dharī‘a*, vol. 21, 253, no. 4896.

⁸³ See e.g. F. Jamil Ragep, “Copernicus and His Islamic Predecessors: Some Historical Remarks”, in: *Filozofski Vestnik* 25/2 (2004), 125–142; I discuss the immediate lettrist-neopythagorean context of this mathematical turn in Matthew Melvin-Koushki, “Powers of One: The Mathematicalization of the Occult Sciences in the High Persianate Tradition”, in: *Intellectual History of the Islamicate World* 5/1 (2017), 127–199. It should be noted, however, that Newton’s project is rather different: where the Samarkand astronomers sought to “free astronomy from philosophy”, as Ragep puts it (i.e., from aristotelian phys-

moreover, by philosopher-scientists of the caliber of Khafri – who similarly produced geomantic treatises. Mention may also be made of Mir Shams al-Din Muḥammad Khabīši Kirmāni, Safavid ṣadr under Shah Muḥammad Khudābanda (r. 985–995/1578–1587), as well as his colleague Jalāl al-Din Muḥammad Munajjim-i Yazdi, discussed below, who both combined expertise in astronomy-astrology and geomancy in a pattern now familiar.⁸⁴ Geomancy, then, was clearly seen in Iran as a mainstream applied mathematical science by the late 9th/15th century, and often pursued as such in a courtly context – including by the sultan-scientist (*al-sultān al-faylasūf*) Ulugh Beg himself.

15. Shams al-Din Muḥammad Khafri (d. 942/1535), *R. dar Raml*

On this work and author see below.

16. Ḥaydar b. Muḥammad Rammāl-i Iṣfahānī, aka Tabrizi (fl. mid-10th/16th c.),
Kashf al-Asrār

Based on the titles he cites in and the general *imamophilic* tenor of this work's introduction, Ḥaydar b. Muḥammad may be presumed to have flourished in Safavid Iran, where his manual was evidently quite popular (27 MS copies survive in Iran alone). Offering an intriguing glimpse into the career of a professional occultist, he opens the *Kashf al-Asrār* with an autobiographical account of the difficulties he faced in mastering geomancy and other occult sciences. After struggling fruitlessly over a long period, during which he consulted many books and sought out many authorities, one night the would-be geomancer performed a special invocation for guidance to God, Muḥammad, 'Alī and Daniel, together with the appropriate litanies and rituals, then went to sleep; he was accordingly vouchsafed a prognosticative dream. Therein he found himself lost in a vast desert with a bewildering multiplicity of paths, and was unable to tell which would lead to a city. But suddenly he came across a grand caravan led by 'Alī b. Abi Ṭālib himself, the king of saints (*shāh-i awliyā*), who revealed to him the highway to an enormous city; there he traveled, and then awoke. This dream was fulfilled a month later: the author finally found a teacher capable of dispelling all his confusion as to the technical details of geomancy as propounded in the standard works (*kitāb-hā-yi mu'tabar*) on the subject. Here he cites the following titles: the *Shajara*, the *Miṣbāḥ*, the *Miftāḥ al-Mafātīḥ*, the *Shāmīl*, the *Kāmīl*, *Khulāṣat al-Baḥrayn* and the *Kashshāf*, together with the untitled treatises (sg. *zubda*) of Mullā Mu'in and Shams al-Din Muḥammad Khafri – the curriculum of a Safavid geomancer-in-training.⁸⁵

ics), the Englishman mathematized celestial physics and dynamics (my thanks to Sonja Brentjes for this observation).

⁸⁴ Iskandar Beg Munshi, *Tārīkh-i 'Ālam-ārā-yi 'Abbāsī*, ed. Īraj Afshār, 2 vols., 3rd ed., Tehran: Amir Kabir 1382 Sh./2003, 321.

⁸⁵ MS Majlis 12593/2, ff. 35b–36b; this copy was made in 1078/1668 (f. 81a), the *terminus ad quem* for the composition of this treatise.

While no biographical information is extant on this Isfahani-Tabrizi geomancer, his citing of Rūyānī's and Khafri's works in particular, and his valorization of 'Alī as the *shāh-i awliyā*, suggests a composition date in the early to middle decades of the 10th/16th century. It also suggests that Ḥaydar b. Muḥammad is to be identified with one Ḥaydar b. Muḥammad Munajjim, author of the geomantic work *Zubdat al-Dhakhbā'ir fī Istikbrāj al-Ḍamā'ir* (which, like Hidāyat Allāh, identifies distinct Berber, Maghribi, Egyptian, Indian and Rūmi geomantic methodologies),⁸⁶ as well as, perhaps, a *Tuhfa-yi Shāhī* on astronomy (MS Malik 3214/13). The same individual also copied a *majmū'a* (MS Raḍavi 5632) comprising works on astronomy-astrology and geomancy, including the *Rawḍat al-Munajjimin* of Shahmardān b. Abi l-Khayr Rāzi (fl. 477/1084) and an *Asrār al-Raml* consisting of extracts from al-Zanāti's work by the same title.

But most intriguingly, the Tabrizi styling and probable Safavid origin of this geomancer suggest the possibility of his identification with that shadowy Ḥaydar Rammāl (d. btw. 966–972/1559–1565) – Shah Ṭahmāsb's (r. 930–984/1524–1576) tutor in Tabriz – who was to become such an influential fixture at the Ottoman court, though this seems unlikely based on our current knowledge of the latter; see below.⁸⁷

17. Sayyid 'Abd Allāh Ḥusaynī Balyānī, aka Shāh Mullā Munajjim-i Shīrāzi (fl. 963–984/1556–1576), *Jabān (Jibāt) al-Raml*

Dānīshpazhūh notes that an early draft of the *Jabān al-Raml* was written in 963/1556 for Shah Ṭahmāsb;⁸⁸ a note at the beginning of one copy of the work states that the author was the Safavid shah's contemporary.⁸⁹ (It will be recalled that Sayyid Balyānī supplied Ṭahmāsb's request for a geomantic manual with a new version of Ṭabasī's *Miftāḥ al-Kunūz*.) This popular treatise, of which at least 15 MS copies survive in Iran alone, appears to have been rededicated to Shah Muḥammad Khudābanda in 984/1576.⁹⁰ Muḥammad 'Alī Mudarris argues that the work was first printed in Bombay under the incorrect title *Jabān al-Raml*,

⁸⁶ Qom, MS Markaz-i Ihyā' 3608/2, 19 ff. (13th/19th c.). Another copy of this work (MS Majlis 130-*fayrūz*, 58 pp. [1248 AH]) attributes it to a Muḥammad b. 'Alī Iṣfahānī.

⁸⁷ While such an identification is further supported by Ḥaydar Rammāl's association in Ottoman sources with Karaca Dağ, just to the west of Tabriz (Cornell Fleischer, "Seer to the Sultan: Haydar-i Remmal and Sultan Süleyman", in: *Cultural Horizons: A Festschrift in Honor of Talat S. Halman*, J. Warner, ed., Istanbul and Syracuse: Syracuse Univ. Press, 2001, vol. 1, 290–299, esp. 296–297), the same placename could refer to Montenegro, especially given Ḥaydar's possession of a *sancak* in Macedonia. More problematic is the question of style: the Persian prose of the *Kashf al-Asrār* is considerably more polished and erudite than that of Ḥaydar's one known Persian work, *Mir'āt al-Karwāyn*, presented to Sultan Süleymān, which suffers from grammatical infelicities (my thanks to Professor Fleischer for these observations).

⁸⁸ Dānīshpazhūh, "Yak Parda", 992.

⁸⁹ MS Majlis 2782, 6.

⁹⁰ Thus e.g. MS Majlis 2782.

which has therefore stuck, but the correct title is in fact *Jibāt al-Raml* since the sections are called *jibats* and the subsections *ufuqs*.⁹¹ However, I would suggest that the title *World of Geomancy* seems to better reflect the geographical theme of this work, which features four directions, East, North, West and South, each with multiple horizons. Sayyid Balyāni reports in the introduction that he traveled widely in order to master astrology, geomancy and lettrism, being inspired in Shiraz in 963/1556 by his brother Shams al-Din Muḥammad, an accomplished occultist, with respect to geomancy in particular; he therefore determined to set down (or revise) his findings after 20 years of effort, in 984/1576. He similarly speaks of his brother Shams al-Din as his teacher in a second substantial work on geomancy, titled *Makhzan al-Asrār*, which has three sections to the *Jabān al-Raml*'s four (preserved only in Qom, MS Masjid-i A'zam 2034) – perhaps the first draft of the latter? In the preface to both texts, Sayyid Balyāni states that he therein but expands upon the comprehensive approach of his grandfather 'Abd Allāh Awliyā (d. ca. 900/1494) in his own *Riyāḍ al-Ṭālibin* – a datum repeated by Hidāyat Allāh.

Most notably, the *Jabān al-Raml* represents the first geomantic manual, Persian or Arabic, to attempt a full synthesis of the various schools of geomancy flourishing throughout the Islamicate world. (As of this writing I have been unable to access any manuscript copies of its older model, the *Riyāḍ al-Ṭālibin*, so cannot say to what extent it follows or departs from that work.) It may also be the first to formally insist on the science's status as an application of the *quadrivium* – and hence a scientific tool for studying the cosmos. That its seventh section is devoted to lettrism – shades of Luṭf Allāh Nishāpūri's *Khulāṣat al-Baḥrayn* – is similarly notable in this context.⁹²

18. Hidāyat Allāh Munajjim-i Shirāzi (fl. 962–1001/1555–1593),
Qavā'id al-Hidāya

This Shirazi geomancer was Sayyid Balyāni's most successful student, and responsible for propagating his teacher's comprehensive synthetic system in Mughal India; I will therefore focus closely on his *Qavā'id al-Hidāya* in what follows, for as a recasting and expansion of the *Jabān al-Raml* it represents the most comprehensive picture of the state of the art throughout the Persianate world in the late 10th/16th century. There are at least seven surviving MS copies in Iran, the earliest two copied in 1003/1595, and include one in the luxury geomantic *majmū'a* MS Majlis 12563, my source here; many more may be presumed to survive in India.⁹³

⁹¹ *Rayḥānat al-Adab fī Tarājīm al-Ma'rūfīn bi-l-Kunya aw al-Laqaḥ*, 8 vols., Tehran: Khayyām, 1369 Sh./1990, vol. 3, 174.

⁹² Storey, *Persian Literature*, vol. 2.3, 482, no. 851, 484, no. 856 (9).

⁹³ See Storey, *Persian Literature*, vol. 2.3, 481, no. 850 (2, 4, 5); Āqā Buzurg, *al-Dharī'a*, vol. 17, 195, no. 1034.

19. Jalāl al-Dīn Muḥammad Munajjim-i Yazdī (d. 1028/1619),
Khulāṣat al-Raml

Also the author of the important chronicle *Tārīkh-i ‘Abbāsī* (aka *Rūznāma-yi Mullā Jalāl*) and *Tuhfat al-Munajjimīn*, an astrological manual, Jalāl al-Dīn Munajjim-i Yazdī was court astronomer-astrologer to Shah ‘Abbās (r. 995–1038/1587–1629) and famously advised his patron to abdicate the throne for three days to manage the Nuqtavi challenge.⁹⁴ Jalāl al-Dīn wrote the *Khulāṣat al-Raml* for Khan Aḥmad Ḥusaynī, ruler of Gilan (r. 943–975/1536–1568, 985–1001/1577–1593). Seven MS copies survive in Iran; another work with this title by Sayyid Turāb-‘Alī was printed in lithograph (India, 1854).⁹⁵

* * *

Such, then, are some of the most influential geomancers of the Persianate world through the turn of the 11th/17th century; together they offer a snapshot of the progressive geographical spread and epistemological and technical development of our divinatory science from the Ilkhanid period onward. For later Persian geomancers, moreover, the intellectual prestige of these authorities and their peers remained wholly intact. Thus Muḥammad Bāqir Ṭabāṭabā‘ī Yazdī, in compiling his own manual in 13th/19th-century Damascus, depends heavily on the *Miṣbāḥ* and al-Zanāti’s *Fuṣūl*, followed by Tūṣī’s *Vāfi*, Hidāyat Allāh’s *Qawā‘id al-Hidāya*, the *R. Surkhāb* and Nāṣir b. Ḥaydar’s *Irshād al-Raml*; he also quotes Ḥasan Abarqūhī’s *ur-jūza*.⁹⁶ In one place Muḥammad Bāqir gives a representative list of authorities as follows, testifying to the final fusion of western and eastern geomantic traditions: the author of the *Kashshāf*,⁹⁷ Waḍ‘ (?) al-Dīn al-Maghribī, Afḍal al-Dīn al-Rūmī,⁹⁸ Māshā‘allāh al-Miṣrī,⁹⁹ Sulṭān al-Muḥaqqiqīn Nāṣir al-Dīn al-Ṭūṣī, al-Imām al-

⁹⁴ See e.g. Kathryn Babayan, *Mystics, Monarchs, and Messiahs: Cultural Landscapes of Early Modern Iran*, Cambridge, Mass.: Harvard Univ. Press, 2002, 350–351; ‘Alī Riḍā Rayḥān, *Āyina-yi Dānistvarān*, Qom: Kitābkhāna-yi Mar‘ashī, 1372 Sh./1993, 661–662; ‘Alī Aṣghar Muṣaddīq, “Khāndān-i Munajjim-i Yazdī u Tārīkh-nigāri-yi Dawra-yi Ṣafavī”, in: *Kitāb-māb-i Tārīkh u Jughrāfiyā* 37–38 (Adhar 1379 Sh./2000), 4–7.

⁹⁵ Storey, *Persian Literature*, vol. 2.3, 481, no. 849; copies include MSS Majlis 12611/2 and 12543; MSS Mar‘ashī 11532/1 and 6393; and MS Mawlavī Muḥammad Shafī‘ 239/325 (Lahore).

⁹⁶ *Nafahāt al-Asrār*, 29.

⁹⁷ Not known to be a title of a geomantic work – perhaps Ḥaydar b. Muḥammad Rammāl-i Iṣfahānī’s *Kashf al-Asrār* is meant here?

⁹⁸ I.e., Afḍal al-Dīn Abū l-Qāsim b. ‘Abd al-Ṣamad b. ‘Abd al-Jalīl (or: ‘Abd al-Malik b. ‘Abd al-Khalīl) al-Rūmī, author of the *Shāmil al-Ḥuṣūl*, or possibly the *Kāmil al-Ḥuṣūl*, assuming these are different works; they may also be a single work whose full title is *Shāmil al-Uṣūl wa-Kāmil al-Ḥuṣūl*.

⁹⁹ I.e., Māshā‘allāh al-Baṣrī (often mistranscribed as al-Miṣrī) (d. 199/815), the famed Jewish astrologer.

Zanāti, ‘Izz al-Dīn al-Abharī,¹⁰⁰ al-Ḥājī Ḥasan al-Abarqūhī and al-Ḥājī Ḥasan al-Ṭabasī.¹⁰¹

But such lists give little sense of the actual *texture* or *sociocultural role* of the Persianate geomantic tradition. For reasons of space I accordingly focus on Hidāyat Allāh Munajjim-i Shirāzi, star student of Sayyid ‘Abd Allāh Balyānī and émigré to India, whose *Qavā'id al-Hidāya*, as noted, serves as epitome of that tradition at the moment of its greatest sophistication and vastest synthesis. Its model, the *Jabān al-Raml*, constructs a conceptual framework for the science that is explicitly “international”: it classifies the regional schools of practice into four broad groups – the Geomantic East (India, Khurasan, Transoxania and points east), the Geomantic North (Fars, Iraq, Gilan), the Geomantic West (the Maghrib, Egypt, Syria, Arabia) and the Geomantic South (the most difficult and occult method, invented by Ṭabasī, exclusive to the saintly elite in all regions) – and provides a full treatment of each. At the same time, Sayyid Balyānī imparts to his work a rich Twelver Shī‘i flavor in substantiation of his Safavid patrons’ total identification of Alidism and occultism.

Hidāyat Allāh, his student and fellow Shirazi, chose rather to seek his fortune in India. (Again, the Shirazi stemming of both men is indicative: that city seems to have functioned as the capital of Persianate occultism during the centuries here in view, and was a major supplier of professional occultists to the neighboring Mughal and Ottoman Empires during the 10th/16th century.)¹⁰² To attract Mughal patronage, in 962/1555 he adapted his teacher’s manual to Mughal imperial purposes and dedicated it to Emperor Akbar in celebration of the latter’s impending accession, then rewrote the work in the month of Farvardin in year 37 of Akbar’s Ilāhi calendar – that is, at the spring equinox of the year 1001 (March 1593):¹⁰³ the moment of dawning of the Islamic millennium over the horizon of Akbar’s sacred body, itself effulging forth the talismanic cosmocrator’s eschatological imperium over all of India. The *Qavā'id al-Hidāya* was thus one of the many works the Mughal emperor commissioned to celebrate the millennium’s arrival, including the feted *Ā'in-i Akbarī* and *Tārīkh-i Alfī*. Hidāyat Allāh’s claim to comprehensiveness and unprecedented clarity in tackling his occult subject, in other words, is na-

¹⁰⁰ Presumably to be identified with the commentator on Naṣir al-Dīn Ṭūsī’s geomantic treatise, Nūr al-Dīn Faṭḥ Allāh Abharī (fl. 786/1384); see the entry on Ṭūsī above.

¹⁰¹ *Nafahāt al-Asrār*, 42. Ṭabāṭabā’ī Yazdī also references Khwāja Abū l-Ma‘ālī l-Nisābūrī’s *R. Miṣṭāḥ al-Mushkilāt* (28), Afḍal al-Dīn Abū l-Qāsim al-Rūmī (identified as a leading authority) (29), Abū Yūsuf Aḥmad al-Maghribī’s *al-Jāmi‘* (38) and Abū Naṣr al-Farāhī (43).

¹⁰² On this occultist diaspora see Matthew Melvin-Koushki, “The Occult Sciences in Safavid Iran and Safavid Occultists Abroad”, in: *The Safavid World*, Rudi Matthee, ed., New York: Routledge, 2018 (forthcoming).

¹⁰³ *Qavā'id al-Hidāya*, MS Majlis 12563/4, p. 316. The *Qavā'id al-Hidāya* was first written for the emperor in 962/1555 just before his accession and reworked in 1001/1593 (Storey, *Persian Literature*, vol. 2.3, 482, no. 850 [4, 5]). Another geomantic work, the *Shajara u Thamarā* cited by Hidāyat Allāh, was also copied for Akbar at the instance of one Kamāl al-Dīn Ḥasan (Storey, *Persian Literature*, vol. 2.3, 487, no. 856 [44, 45]).

kedly calculated to reflect and perform his patron's own claims to intellectual-spiritual-political universality and manifestness.¹⁰⁴ The Shirazi geomancer, that is, is to be considered a member of that group of elite scholars described by Abū l-Faḍl 'Allāmī (d. 1011/1602) as being those

who in their stellar brilliance comprehend mysteries both occult and manifest, possessing full command of the two realms by virtue of exalted nature and ample aptitude; yet they account themselves mere passive recipients of the effluxion that emanates from the imperial court.¹⁰⁵

Equally significant in this context is the fact that *Hidāyat Allāh* replaces the encomium to 'Alī that figures in the preface to his teacher's work with a poem heavily astrological in its symbolism, a design feature sure to catch the fancy of Akbar, who had made planetary worship the basis of his daily court ritual – precisely as a means of sidestepping exclusivist Safavid claims to 'Alī and the *walāya* of the Imams.¹⁰⁶

Nor was the readership of *Hidāyat Allāh*'s geomantic manual confined to Akbar's court; that it can be presumed to have had a ready and eager market among persophone Indian elites more generally is indicated by no less an authority than Abū l-Faḍl, chief architect of Akbar's new Indo-Timurid imperial culture, who explicitly lists the science as part of his ideal core curriculum for all students in Akbar's realm, to be studied even before medicine or logic:

The following sciences should be acquired consecutively: ethics, arithmetic, accounts, agriculture, mensuration, geometry, astrology (*nujūm*), geomancy (*raml*), economics, government, medicine and logic, then [the rest of] the natural and mathematical sciences and metaphysics, then history.¹⁰⁷

As the most comprehensive and systematic Persian geomantic manual circulating in the later 10th/16th century, then, the *Qavā'id al-Hidāya* slotted perfectly into this new imperial curriculum. And despite first being developed for *Safavid* imperial purposes, its universalist approach to the science was likewise in perfect consonance with Akbar's famed, radically ecumenical doctrine of Universal Harmony (*ṣulḥ-i kull*), as well as his claims to talismanic divinity. This fact alone indicates that a higher degree of continuity obtained between competing Safavid and Mughal

¹⁰⁴ Specifically, he asserts that what "had been written in books with extreme laboriousness and allusiveness has now been fully clarified by *Hidāyat Allāh*, in the event anyone's home"; *Qavā'id al-Hidāya*, MS Majlis 12563/4, 373.

¹⁰⁵ Abū l-Faḍl 'Allāmī, *Ā'in-i Akbarī*, H. Blochmann, ed., 12 vols., Calcutta: Asiatic Society of Bengal, 1869–1872, vol. 1, 232.

¹⁰⁶ See Azfar Moin, *The Millennial Sovereign: Sacred Kingship and Sainthood in Islam*, New York: Columbia Univ. Press, 2012; Audrey Truschke, "Translating the Solar Cosmology of Sacred Kingship", in: *Medieval History Journal* 19/1 (2016), 136–141; Matthew Melvin-Koushki, "Early Modern Islamicate Empire: New Forms of Religiopolitical Legitimacy", in: Armando Salvatore, Roberto Tottoli and Babak Rahimi, eds., *The Wiley-Blackwell History of Islam*, Malden, Mass.: Wiley-Blackwell, 2017, 353–375.

¹⁰⁷ *Ā'in-i Akbarī*, vol. 1, 202.

imperial ideologies than is usually acknowledged: through the simple expedient of replacing 'Alī with astrology in its proem, a Safavid geomantic manual could and did serve as prop to Mughal imperialism essentially unchanged.¹⁰⁸

In his recent pioneering study on the performance of sacral kingship in Mughal India, Azfar Moin dwells frequently on the various Mughal dynasts' dependence on occult scientists, particularly astrologers, lettrists and geomancers, for corroboration of their claims to human perfection and eschatological imperialism.¹⁰⁹ This extends to the Subcontinent the findings of Cornell Fleischer and Kathryn Babayan, who have definitively shown the same for the Ottoman and the Safavid courts respectively.¹¹⁰ Most crucially for our purposes here, Fleischer's studies of yet another Safavid geomancer, Ḥaydar Rammāl, and his long, brilliant career at Sultan Süleymān's court have made it abundantly clear that Ottoman imperial ideology in the 10th/16th century – and by extension early modern Turko-Mongol Perso-Islamic imperial ideologies as an interactive set – simply cannot be understood without reference to geomancy specifically and the occult sciences generally.¹¹¹ (Attention to the question of occultism at the Uzbek court remains a desideratum.¹¹²) In an important aside, Fleischer notes that Ḥaydar the Geomancer influenced Shah Ṭahmāsb just as deeply before decamping to the Ottoman court:

So much did Tahmasb envy Haydar's art, and also fear his own dependence on the Geomancer, that he had Haydar tutor him for six years. Apparently the lessons had a profound and lasting effect on Shah Tahmasb, for when Vincenzo degli Alessandri visited the Safavid court of Qazvin some forty years later, he found that the shah had not left his palace for ten years, preferring to seclude himself with his women and constantly cast geomantic figures; when his prognostications might happen to come true, the women lavished praises on him as God's chosen, the recipient of divine revelation, a true prophet of God.¹¹³

While somewhat extreme, Ṭahmāsb's singleminded commitment to the Danielic art provided a hospitable framework for the production of such landmark works

¹⁰⁸ On the smooth Irani-Hindustani (and Turani) continuities defining High Persianate culture more generally see Mana Kia, "Contours of Persianate Community, 1722-1835", PhD dissertation, Harvard Univ., 2011.

¹⁰⁹ Moin, *The Millennial Sovereign*.

¹¹⁰ Babayan, *Mystics, Monarchs, and Messiahs*.

¹¹¹ Cornell H. Fleischer, "Ancient Wisdom and New Sciences: Prophecies at the Ottoman Court in the Fifteenth and Early Sixteenth Centuries", in: *Fahnama: The Book of Omens*, Massumeh Farhad and Serpil Bağcı, eds., London: Thames & Hudson, 2009, 232-243, 329-330; Cornell H. Fleischer, "Shadow of Shadows: Prophecy in Politics in 1530s Istanbul", in: *International Journal of Turkish Studies* 13 (2007), 51-62; Cornell H. Fleischer, "Seer to the Sultan".

¹¹² On the *Kashf al-Shajara* of Maḥmūd Kawtharī b. Ismā'īl, geomancer to Muḥammad Shībānī Khan, see n. 55 above.

¹¹³ Fleischer, "Shadow of Shadows", 60. Again, there is the possibility, though slim, that this geomancer is to be identified with the Ḥaydar b. Muḥammad Rammāl-i Iṣfahānī-Tabrizī listed above; if so, the list of works the latter gives in his *Kashf al-Asrār* thus represents the geomantic curriculum Ṭahmāsb likely studied under his and Sayyid Balyānī's tutelage.

as Sayyid Balyāni's *World of Geomancy*, which synthesized the Persianate geomantic tradition into a cohesive whole – a feature that made it attractive to Safavid, Mughal and Ottoman sovereigns in equal measure. Nor was such enthusiasm limited to the ruling elite; as Jean Chardin (d. 1125/1713), Huguenot sojourner in Safavid Iran under Shah Sulaymān (r. 1077–1105/1666–94), patronizingly observed of Safavid society a century later: “[M]ost [Persians] are dependent on astrologers and other diviners as a child is dependent on its nurse.”¹¹⁴

I do not propose to further examine the political applications of geomancy here, but merely to pack a little flesh onto the bare bones of this new narrative of Islamicate early modernity by supplying specifics as to how the science was ranked and practiced.¹¹⁵ Again, Hidāyat Allāh's geomantic manual is a worthy place to start toward this end, since it is an exceptional specimen of the genre and was doubtlessly widely read, perhaps even at the Ottoman court. (Hidāyat Allāh's elder contemporary Ḥaydar Rammāl, who produced annual prognostications in his official capacity as court geomancer, seems not to have written a manual of his own while in Ottoman employ.¹¹⁶) More importantly, it represents the state of the art and its canon at the turn of the 11th/17th century: like Ṭabāṭabā'i Yazdi, Hidāyat Allāh provides a list, the source for the later author's, of the 50 titles his work effectively summarizes.¹¹⁷

Writing from his privileged vantage point at the dawn of the Islamic millennium, then, how does our Shirazi émigré rationalize and summarize the preceding four centuries of geomantic theory and practice? He begins by restating his teacher's schema as follows:

Know (God aid you) that this noble science is pursued in every corner of the earth, with four main schools of practice. Indians, Khurasanis, Transoxanians and easterners gener-

¹¹⁴ Quoted in Massé, *Persian Beliefs*, 240.

¹¹⁵ On the political applications of the occult sciences in the early modern Persianate world see Matthew Melvin-Koushki, “Astrology, Lettrism, Geomancy: The Occult-Scientific Methods of Post-Mongol Islamicate Imperialism”, in: *Medieval History Journal* 19/1 (2016), 142–150; and my forthcoming *The Occult Science of Empire in Aqqeyunlu-Safavid Iran: Two Shirazi Lettrists*.

¹¹⁶ See Fleischer, “Seer to the Sultan”, 296; one such annual prognostication is preserved in the Topkapı Library collection as MS Hazine 1697. If, however, he is to be identified with our Ḥaydar b. Muḥammad, then he was responsible for the *Kashf al-Asrār*, one of the most popular Persian geomantic manuals ever produced.

¹¹⁷ See n. 51 above for a synthetic list combining the two. The list as it appears in the *Qavā'id al-Hidāya* is as follows (MS Majlis 12563/4, p. 323): *Miftāḥ al-Kunūz*, *Kanz al-Daqa'iq*, *Shajara u Thamara*, *Shāmil al-Huṣūl*, *Tuhfa-yi Shābi*, *Uṣūl al-Raml*, *ʿAyn al-Raml*, *Shamʿ al-Raml*, *Miftāḥ-i Maṣātib*, *Vāfi*, *Si Bāb*, *Fuṣūl*, *Abwāb-i Anvār*, [the work of] Uqlidi, [the work of] Ṭarābulsi, [the work of] Rawshani, the treatise of Kulah-i Kabūd, *Kashf al-Asrār*, *Talkhiṣ*, the treatise of Abū l-ʿAbbās Būnī, *Minhāj al-Asrār*, *Jāmiʿ al-Kifāya*, *Khulāṣa*, *Khulāṣat al-Baḥrayn*, *Tabdih*, *Dbakhira*, *Zubda*, *Natijat al-ʿUlūm*, *Qavā'id*, *Nuzhat al-ʿUqūl*, *Jāmiʿ*, the *Kāmil* of Ḥusayn-i Qaffāl, *Riyāḍ al-Ṭālibin*, *Shajara-yi Awrāq*, *Sāʿir al-Awliyā*, *Mishbah*, *Miftāḥ*, *Hidāyat al-Nuqta*, *R. Surkhāb*, [the work of] Biyābānaki, [the work of] Bābā Kūhi, *Tawdih*, *Takwih*, the work of Mullā Shāh, *Taqvim al-Raml*, *Aṣṭ-i Maṣātib*, *Zayn al-Raml*, *Kāmil al-Huṣūl*, *Jahān al-Raml*, *Jāmiʿ al-Asrār*.

ally rely on the *ABDĤ* cycle and the methods laid out in the *Shajara* [*u Thamara*] in their practice, a school I here term the Geomantic East. Farsis, Iraqis, Gilanis and the like rely on the Occupation (*sakan*) cycle [developed by al-Zanāṭi] and the procedure laid out in the *Mafātīḥ* for deriving their readings, a school I term the Geomantic North. Maghribis, Egyptians, Syrians and all Arabians favor the *BZDĤ* cycle, [established by Luqmān, who named it for his son,] a school I term the Geomantic West. There is a further method, extremely recondite, known as the Most Complete (*aṣaḥḥ*), a school I term the Geomantic South; [it was developed by Ṭabasī in particular], and elite practitioners in every region perform divinations (*kaṣḥf*) and miracles (*karāmāt*) [on its basis]. This includes Sayyid Muḥammad Nūrbakhsh and his like; some credit him with teaching this pristine form of geomancy. My own teacher's grandfather, Awḥad al-Dīn 'Abd Allāh [Balyānī], known as 'Abd Allāh Awliyā (God forgive his sins), composed a work on this last method entitled *Gardens for Seekers* (*Riyād al-Ṭālibīn*), and therein summarized all [four] methods on the basis of the holy Shaykh Ḥājji Hasan Ṭabasī's work (the shaykh being famed for his mastery of *jinn* magic [*taskhīr-i jinn*]) ... This book, then, which I have titled *Methods of Guidance* (*Qavā'id al-Hidāya*), offers a summary treatment of these four methods, with some additional material, to enable anyone who reads and achieves a basic understanding thereof to penetrate people's innermost thoughts.¹¹⁸

The various regional schools of geomantic thought are therefore to be distinguished by the 'cycles' (sg. *dā'ira*) they prefer to employ. A cycle, Hidāyat Allāh explains, is simply a specific sequence (*tartīb*) of the 16 geomantic figures, with each sequence featuring different types of correspondences – elemental, humoral, temporal, astrological, lettrist, etc. And the four cycles he identifies above are far from the only ones in common use. The geomancer has at his disposal a rather larger number of cycles; which he employs in any given reading depends on the nature of the information being sought and the degree of detail required. Hidāyat Allāh lists seven by way of example:

1. The Occupation (*sakan*) cycle is the most basic; it begins with Laetitia and ends with Via.
2. The Constitution (*mizāj*) cycle tells the querent what day he can expect to realize his desire; it begins with Acquisitio and ends with Cauda Draconis.
3. The *BZDĤ* cycle, aka the Number (*ʿadad*) cycle, is used to tell periods of time; it begins with Puer and ends with Populus.
4. The Letter (*ḥarf*) cycle is used to reveal names (a very popular application of the science); it too begins with Laetitia and ends with Via, with the first twelve figures being assigned two letters each and the last four figures only one.
5. The Arabic Alphabet (*alḥjad-i ʿarabī*) cycle, aka the Element (*ʿunṣur*) or Interior (*dākhil*) cycle; it begins with Laetitia and ends with Populus.
6. The *ABDĤ* cycle, aka the Element (*ʿunṣur*) or Arabic Alphabet (*alḥjad-i ʿarabī*) cycle, which is also popular; it too begins with Laetitia and ends with Populus.

¹¹⁸ *Qavā'id al-Hidāya*, MS Majlis 12563/4, pp. 315–316; I supplement this passage on the basis of pp. 374, 427, 456.

7. The Most Complete (*aṣaḥḥ*) cycle, which has a different organizing principle and procedure; it begins with Laetitia and ends with Via.¹¹⁹

Such cycles, moreover, are typically combined in a standard geomantic reading to provide the fullest picture possible in response to a question:

Know that all geomantic readings (*aḥkām*) as to a person's innermost thoughts (*ḍamīr*), a judgment (*ḥukm*), a *jinn*, a name, a buried object (*ḍafīn*), a number or a period of time are derived from these cycles (*davāyir*). And the cycles that are here laid out may be utilized [simultaneously]. For example, say you obtain a reading (*ḥukm*) based on the Occupation cycle; you then find [information about] number and period from the Number cycle, the name from the Letter cycle, the relevant day from the Constitution cycle, the fruit that will be mentioned from the Fruits (*ḥavākīl*) cycle, [and so on]. Every datum, [in short,] must be sought in its associated cycle.¹²⁰

The upshot: geomancy offers the possibility of universal knowledge about events, ideas, entities, identities, objects and locations at the most granular level – even *fruits* can be captured in the trammel of its binary code. This stirring prospect, of obvious utility and appeal to messianic sovereigns and saints, openly invites the geomancer and his royal patron to partake in prophethood. Geomantic tradition associates the science with up to seven prophets in particular:

It is commonly accepted among the practitioners of this art that the science of geomancy is attributable to seven prophets: first, Adam, God's Pure One (*Ṣaḥī Allāh*); second, Seth; third, Enoch (*Idrīs*); fourth, Jeremiah; fifth, Isaiah; sixth, Luqmān; seventh, Daniel.¹²¹

More specifically, origin stories for geomancy assign it the role of heaven-sent prophetic miracle (*muʿjiza*) that each prophet requires to corroborate his claim: magic for Moses, healing for Jesus, the inimitable Quran for Muḥammad – and geomancy for Daniel. (It should here be noted that the explosion in popularity of the Daniel apocalypses (sg. *malḥamat Dāniyāl*) during the 10th/16th century, particularly among Ottoman elites, is intimately connected to his geomantically-based status as prognosticator *par excellence*.¹²²) In Hidāyat Allāh's rendition of the classic Daniel narrative, emphasis is placed on the miraculous level of detail that the prophet is able to provide his clients as a professional geomancer, as well as the conversion of the *entire world* to Islam by means of this occult art:

¹¹⁹ *Qavā'id al-Hidāya*, MS Majlis 12563/4, pp. 332–333 (paraphrased).

¹²⁰ *Qavā'id al-Hidāya*, MS Majlis 12563/4, p. 456.

¹²¹ *Qavā'id al-Hidāya*, MS Majlis 12563/4, p. 313. Hājji Khalifa reports that the seminal *Miṣbāḥ al-Raml* gives a similar list of six prophets in which Seth is not included (*Kashf al-Zunūn*, vol. 1, 912).

¹²² Cornell Fleischer, "Lawgiver as Messiah: The Making of the Imperial Image in the Reign of Süleyman", in: Gilles Veinstein, ed., *Soliman le magnifique et son temps*, Paris: École du Louvre, 1992, 159–177: 170. On the *malāḥim* genre generally see Mohammad Masad, *The Medieval Islamic Apocalyptic Tradition: Divination, Prophecy and the End of Time in the 13th Century Eastern Mediterranean*, PhD dissertation, Washington Univ., St. Louis, 2008.

Adam the Pure (peace be upon him) was the first person in the world who wished to know his status (*ḥāl*) and that of his children and discover where each one was, and Gabriel taught him [geomancy to this end]. This science was then transmitted from him to the abovementioned prophet, [viz., Seth or Enoch,] after which a long age passed during which it remained obscure [and unknown] to the people of the world until such time as the Lord of Glory revealed it by way of prophetic inspiration to Daniel (peace be upon him). [This prophet] had announced his messengerhood, but since he performed no prophetic miracle (*muʿjiza*) [in support of his claim] his enemies and opponents sought to have him killed. Shamʿūnyā¹²³ (peace be upon him) accordingly fled, and he wandered in the deserts and wastes for three days and nights until weakness and hunger overcame him. He cried out to the Court of Oneness (*dargāb-i aḥadiyyat*), whereupon Gabriel descended and taught him this science by drawing four dots in the sand (or, as some say, one dot with four corners), instructing him to the effect that each dot (or each corner) represents the central node of an element (*ʿunṣur*). That is, the first dot is fire, the second air, the third water, the fourth earth. After fully learning this science from Gabriel, Shamʿūnyā too drew several dots in the sand. He then departed the sandy waste and returned to civilization, taking up residence in a corner of the Egyptian bazaar and keeping some sand nearby.

[One day] a shepherd approached to ask about his status. [The prophet] replied: “Your name is Surkhāb, and you have 1,470 sheep, three head of mountain goat, seven head of cow and 15 head of donkey. You also have a brother who is on a journey, and in the place he now is he has fallen in love with a Khotanese girl named Ṭighān.” Later a merchant came to ask about his missing son. [The prophet] replied: “Your son was trysting with the daughter of someone named Aḥmad in a place of ruins, and when her father found out he killed both of them by throwing them down a well.” [...] ¹²⁴ When news of this [act of divination] reached the king, he summoned Shamʿūnyā and went with him to the well in question and brought out the corpses, whereupon the people believed in [Daniel] mightily

[Daniel] performed such prophetic miracles until a number of notables at court, including the king’s own son, [sought to] learn this science from Daniel (peace be upon him), who thus became the king’s constant companion. One day the king was in an expansive mood and called a private meeting with this same group of notables in which they discussed geomancy and the miracles wrought by earlier prophets. The king then posed this question: “Is there a prophet in the world today?” The group of them having focused their intention on this question, they performed a geomantic reading. The result: “There is indeed a prophet, and he is in our kingdom, nay, in our very garden and in attendance at this very session; he is our teacher and his name is Shamʿūnyā, and his appearance is thus.” The prophet in question was of course the one present, and accordingly everyone professed belief in him and became Muslims. They also summoned the people of Egypt to behold these prophetic miracles, whereupon they too became Muslims, and spread the news of [Daniel’s] prophethood throughout the world until all believed.¹²⁵

¹²³ As *Hidāyat Allāh* later explains, Shamʿūnyā was the prophet’s original name, but after mastering geomancy he adopted the name Daniel to numerologically accord with this science (*Qavāʿid al-Hidāya*, MS Majlis 12563/4, p. 322).

¹²⁴ A reconstruction and full analysis of these geomantic readings here intervenes.

¹²⁵ *Qavāʿid al-Hidāya*, MS Majlis 12563/4, pp. 316–322.

Geomancy as mathematical-natural occult science

The ease and fluency with which Hidāyat Allāh Munajjim was able to depict the universality of geomancy suggest its vitality and cultural prestige during the 7th–10th/13th–16th centuries, particularly in the Persianate east. As the epigram at the beginning of this article testifies, Ibn Khaldūn (d. 808/1406) himself, despite his dedicated though confused anti-occultism, was constrained to admit the prevalence of geomantic practice “in all civilized lands”¹²⁶ – an enviable status that, by all accounts, only improved after the 8th/14th century. How, then, to explain its ubiquity among scholarly and ruling elites over the course of many centuries?

It must first be noted that geomancy was increasingly reclassified as a *mathematical* science from the 5th/11th century onward – a simple but oft-forgotten fact that both reflects and explains its enduring appeal to philosophers, astronomers and mathematicians, including in the first place prominent thinkers like Naṣīr al-Dīn Ṭūsī and Shams al-Dīn Khafri, as well as lesser lights like ‘Abd al-Qādir Rūyānī.¹²⁷ This appeal, in turn, signals its importance to history of science more generally. Most significantly, the mathematicalization of geomancy, together with lettrism, was part of a larger trend in Persianate intellectual history that culminated with the fateful mathematization of astronomy in the 9th/15th century by the members of the Samarkand Observatory, and drove the renaissance of neopythagoreanism in Iran through the Safavid period.¹²⁸

This process began with Fakhr al-Dīn Rāzī (d. 606/1202), the great Ash‘ari philosopher-theologian and committed occultist; his *Jāmi‘ al-‘Ulūm* (aka *Sittinī*), the most comprehensive and innovative of the early Persian encyclopedias, is the first to formally elevate geomancy from the natural sciences to the mathematical. Specifically, it follows the standard Avicennan epistemological progression in treating the natural sciences (*tabī‘iyyāt*), the mathematical sciences (*riyādiyyāt*) and metaphysics (*ilāhiyyāt*) in that order, and under the rubric of the mathematical sciences includes geomancy (*raml*), together with geometry (*handasa*), mensuration (*masāha*), mechanics (*jarr al-athqāl*), war machines (*ālāt al-ḥurūb*), Indian arithmetic (*ḥisāb al-Hind*), mental calculation (*al-ḥisāb al-baḥā‘i*), algebra (*al-jabr wa l-muqābala*), arithmetic (*arithmātiqī*), magic squares (*a’dād al-vafq*), optics (*manāẓir*), music (*mūsīqī*), astronomy (*ḥay’a*), judicial astrology (*aḥkām [al-nujūm]*) and astral magic (*‘azāyim*). While he devotes some five pages to our divinatory art, however, Rāzī’s presentation of geomancy is so rudimentary as to be useless to a would-be practi-

¹²⁶ *Muqaddima*, 108.

¹²⁷ Unlike ‘philosopher’ (*ḥaylasūfī*), ‘astronomer’ and ‘mathematician’ are here used as terms of convenience: the former denotes a specialist in *‘ilm al-bay’a*, while the latter only emerged as a professional term relatively late, and was infrequently used (my thanks to Sonja Brentjes for this observation).

¹²⁸ See Melvin-Koushki, “Powers of One”.

tioner, suggesting that it had not yet entered the intellectual mainstream in the Persianate world.¹²⁹

The *Nafāyis al-Funūn fi 'Arāyis al-'Uyūn* of Shams al-Dīn Āmulī (d. 753/1352), sometime *mudarris* at Sultaniyya under Öljeytū (r. 693–706/1294–1307), represents a second watershed in the Islamicate encyclopedic tradition. It is the most extensive and polished *taṣnīf al-'ulūm* work produced in either Arabic or Persian up to the mid-8th/14th century, and as such came to serve as the primary model for subsequent Persian encyclopedias.¹³⁰ The *Nafāyis al-Funūn* has as its own model Rāzī's *Jāmi' al-'Ulūm*, but it is far more complete and organized than that pioneering work. Most significantly, Āmulī, unlike Rāzī, questions the standard Avicennan hierarchy of sciences so forcefully reiterated a century prior by Quṭb al-Dīn Shīrāzī (d. 710/1311) in his *Durrat al-Tāj li-Ghurra al-Dabbāj*, at times even challenging the latter directly by counterposing what he asserts to be a more consistent classification system and moving toward a new conception of knowledge.¹³¹ At the same time, he concurs with the Maragha philosopher-astronomer in classifying all occult sciences, including alchemy (*kīmīyā*), letter magic (*simīyā*), oneiromancy (*ta'bir*), physiognomy (*firāsāt*) and astrology, as derivative natural sciences – with one pointed and telling exception: geomancy. While Rāzī was the first encyclopedist to treat this form of “terrestrial astrology” as an applied mathematical science (i.e., an instance of the *furū'-i ri'yādī*) associated with astronomy, then, Āmulī goes further: he dissociates the science from astrology on the one hand and all other divinatory sciences on the other.¹³² Though his motive for making it is not clear, Āmulī's move here may be considered an index of geomancy's increase in prestige in Iran during the Ilkhanid period. Moreover, at 20 pages Āmulī's discussion of geomancy is considerably fuller than his model's and detailed enough to serve as a basis for elementary practice.

The prestige of geomancy only continued to increase in the Persianate world from this point forward, such that by the 10th/16th century geomancy was considered an essential and uncontroversial application of the *quadrivium* in its Avicennan formulation. Hidāyat Allāh Shīrāzī, writing some two and a half centuries later, programmatically states:

Know (God guide you) that philosophy (*ḥikmat*) is divided into three categories: the natural sciences (*'ilm-i ṭabī'ī*), the mathematical sciences (*'ilm-i ri'yādī*) and metaphysics (*'ilm-i ilāhī*). The principal mathematical sciences are four: astronomy (*ḥay'at*), arithmetic

¹²⁹ *Jāmi' al-'Ulūm* ('Sittinī'), Sayyid 'Alī Āl-i Dāvūd, ed., Tehran: Thurayyā 1382 Sh./2003, 432–437. In this list geomancy, the 51st science, follows judicial astrology and precedes astral magic.

¹³⁰ Živa Vesel, *Les encyclopédies persanes: essai de typologie et de classification des sciences*, Paris: Institut Français de Recherche en Iran 1986, 39–42.

¹³¹ Vesel, *Les encyclopédies persanes*, 41; Melvin-Koushki, “Powers of One”.

¹³² *Nafāyis al-Funūn*, vol. 3, 537–556. Here again I use the designation “applied mathematical science” as a term of convenience; it is not to be understood in its strict 19th-century European sense.

(*ʿadad*), geometry (*handasa*) and music (*mūsīqī*). The mathematical disciplines derived from these are numerous, and include geomancy (*raml*) and astrology (*nujūm*), which are exceedingly noble and useful sciences.¹³³

At the same time, geomancy's central focus on the four elements – fire, air, water, earth – also gave it a strong claim to being a natural science. As the *Shajara u Thamara*, a popular Persian geomantic manual of the period (formally a commentary on a Persian translation of al-Zanāti's *Shajara*), warns:

The modality of this science is a difficult one, for it is a physiological science (*ʿilm-i ṭabāyīʿī*); whoever does not have a firm grasp of physiology will not be able to use this science.¹³⁴

Our divinatory science, in other words, occupied the epistemological intersection between astrology and alchemy.¹³⁵

But geomancy's status as 'the science of the dot,' which allowed precisely for its dual mathematical-natural status, was not entirely unproblematic: the conception of the four-cornered *nuqta* as primordial embodiment of the four elements emerged as a site of doctrinal controversy in the early 9th/15th century, primarily due to claims put forward by the anarchic Hurufis of western Iran. In particular, it was central to Maḥmūd Pasikhāni's (d. 831/1428) reformulation of the teachings of his erstwhile teacher Faḍl Allāh Astarābādī (d. 796/1394), and drove what was seen by critics to be the thoroughgoing *naturalism* or *materialism* of Nuqtavi doctrine.¹³⁶ As a consequence, contemporary manuals like the *Shajara u Thamara*

¹³³ *Qavā'id al-Hidāya*, MS Majlis 12563/4, p. 312. See also Sonja Brentjes, "The Mathematical Sciences in Safavid Iran: Questions and Perspectives", in: *Muslim Cultures in the Indo-Iranian World During the Early Modern and Modern Periods*, F. Speziale and D. Hermann, eds., Berlin: Schwarz, 2010, 1–77.

¹³⁴ MS Majlis 12563/7, p. 820. The *Thamara-yi Shajara* (presumably identical to the *Shajara u Thamara*) was done into Persian verse in 1263/1847 by one Ashraf b. Muḥammad 'Alī Mashhadī, takhalluṣ Ashraf, a minor poet with occultist proclivities (Aḥmad Ḥusaynī Ishkavari, *Tadbkira-yi Shā'irān-i Fārsī-sarā*, Qom: Majma'-i Dhakhāyir-i Islāmi, 1385 Sh./2006, 29). The dual status of geomancy as both a mathematical and a natural science is further suggested by the existence of *majmū'as* like MS Majlis 12574, which contains the popular Persian geomantic treatise *Sā'ir al-Awliyā*, a Persian medical text (versified instructions for the treatment of various diseases) and a translated excerpt from the Ikhwān al-Ṣafā' on a section of Aristotle's *Physics*.

¹³⁵ Klein-Franke, "The Geomancy of Aḥmad b. 'Alī Zunbul", 33, 35.

¹³⁶ See e.g. Ṣādiq Kiyā's classic study of the movement, *Nuqtaviyān yā Pasikhāniyān*, Tehran: Anjuman-i Īrān, 1320 Sh./1941. In his *Durr al-Yatim*, for instance, Muḥammad Dihdār Shirāzī (d. 1016/1607), émigré to Mughal India and son of the important Safavid lettrist author Maḥmūd Dihdār Shirāzī (fl. 984/1576), classes the Nuqtavis as an execrable subset of those he calls the mass of deficient naturalist-materialists (*jumbūr-i ṭabī'īyyīn u nāqīṣān-i dabrīyān*), on a par with sophists, libertines and most of the infidels (*kafara*) of India (*Rasā'il-i Dihdār*, Muḥammad Ḥusayn Akbari Sāvi, ed., Tehran: Nuqta, 1375 Sh./1996, 131), criticizing their doctrine as follows (Dihdār, *Rasā'il-i Dihdār*, 141): "These [people] who associate themselves with the dot (*nuqta*) mean by it the elemental particles (*ajzā-yi 'unṣuri*) that are in the dot; and their sublimest (*haqīqī*) dot is the center of the earth (*markaz-i khāk*). For this reason, whenever their leader [i.e., Maḥmūd Pasikhāni] would

are concerned to carefully define their epistemological position against such perverse materialism:

[The author of the *Shajara*] states that this is a prophetic science, and also that it is called [the science of] the dot (*nuqta*); it must therefore be determined to which dot this refers. According to geometers (*abl-i Uqlidis*), the dot is that which is indivisible, while astronomers use the same term to refer to the center of a circle, as well as a star [or planet]. In the present book, however, what is meant by the dot is the [primordial four-cornered] single/odd point (*fard*), which is to say fire-air-water-earth. In every technical usage of the term, *fard* is one and double/even (*zarv*) is two. In geomancy, *fard* and *zarv* both apply to the dot. Thus while the dot is a singular point without quality (*kayfiyyat*) or addition (*jabr*) or articulation (*makbraj*), and also dimensionless, when you utter it it becomes a letter – *NQTH* – in the same way that *ALF* (*alif*) is a letter, but also a name (*ism*) composed of the three letters *ALF*. It is therefore evident that *nuqta* is being used in this sense as a word, not as a letter (*harf*) (which according to linguists literally means ‘turning away’ [*bar-zadan*]). So know that *nuqta* has four letters whose values when added come to 164; and this number divided by 16 [i.e., the number of geomantic cells] gives a remainder of four – the elements fire, air, water and earth.

It is in this respect that [the author] exhorts us to keep this science from falling into the hands of the ignorant and unworthy: for they are apt to believe that the dot and the figures formed therefrom are causes (*mu'atthir*) in their own right, thereby damaging their faith. But the educated and intelligent are aware that this is a noble science predicated entirely on the divine will – anything else would contravene the dicta *God does what He will* (Q 3:40, 22:18) and *decrees whatsoever He desires* (Q 5:1). Therefore anyone who believes the dot and geomantic figure to be causes becomes an unbeliever (*kāfir*) – we take refuge in God from such! –, and hence a qualified teacher is necessary to prevent perversion of belief. Otherwise one risks bringing down on oneself divine wrath. Now it may happen that someone who has not had a teacher in the science is able to correctly divine the intentions of a querent (*ḍamir-i sāyil*), this by chance; but such an individual is merely a soothsayer (*az jumla-yi kābinān*), that is, one of those who get their information from *jinn*, [not a geomancer].¹³⁷

Geomancy's status as a *mathematical* occult science appears to have become particularly salient by the early 9th/15th century, when it began to be explicitly associated with lettrism (*‘ilm al-ḥurūf*, *‘ilm al-asmā’*), particularly letter magic (*simiyā’*) and letter divination (*jafr*), both based on number theory (*‘ilm al-‘adad*). This trend, alluded to in the passage above, is best exemplified by Luṭf Allāh b. ‘Abd al-Malik

mention his own name in his treatises he would say ‘I who have dirt on my head’ – that is to say, ‘I who am the center of the earth’! [The Nuqtavis aside], however, the science of the dot (*‘ilm-i nuqta*) is a noble science; for the dot refers to the absolute aspect (*ḥaythiyyat-i muṭlaqa*). The treatise *Secrets of the Dot* (*Asrār al-Nuqta*) by Mir Sayyid ‘Alī Hamadānī (may He sanctify his spirit) is well-known [in this connection]. And according to the felicitating speech of the holy Commander of the Faithful, [‘Alī b. Abi Tālib,] on [the subject of] the dot, *Knowledge is a dot multiplied by the ignorant*, here referring to the ultimate reality (*ḥaqīqat al-ḥaqāyiq*) or first entification (*ta‘ayyun-i arval*).”

¹³⁷ MS Majlis 12546/1, ff. 1b3a; MS Majlis 12563/7, pp. 820821. There are some variations in this passage between these two copies, combined in the translation here.

Nishāpūrī Samarqandī's¹³⁸ *Khulāṣat al-Baḥrayn*, written in 812/1409, which treats of geomancy and magic squares (*al-dād-i vafq*) (the two seas of the title) as complementary sciences.¹³⁹ As he states in the work's introduction:

Know (God grant you felicity) that the science of geomancy and [that of] magic squares are two noble sciences that have been continually pursued with the utmost devotion by kings, sultans and amirs [on the one hand] and by philosophers (*ḥukamā*) and scholars (*ulamā*) [on the other]. Proficiency in these sciences gave them knowledge of esoteric matters (*rumūz*), hidden treasures (*kunūz*), people's innermost thoughts (*sarāyir u ḍamāyir*) and the active properties of things (*khavāṣṣ-i ashyā*); they thereby attained to high estates indeed.

Now it will be asked, why treat geomancy and magic squares together in a single book, given that each science has been treated separately in [numerous] books – how are they related (*munāsabat*)? I answer: [In the first place], it is perfectly acceptable to treat even a hundred sciences in a single book. And as sciences geomancy and magic squares are especially closely related, for the first delivers prognostications (*ḥukm*) and reveals inner states (*ḍamir*) while the second activates the properties [of things] (*khavāṣṣ*) and confers magical control [over them] (*tadbir*). [Beyond these virtues], geomancy and magic squares possess a very many properties (*ahkām, khavāṣṣ*) [that are mutually complementary]. Moreover, the [geomantic] figures are called evens (*zawjān*) or odds (*fardān*), and fivers (*khumāsī*), sixers (*sudāsī*), seveners (*subā'i*) or eighters (*thumāni*), [much like magic squares];¹⁴⁰ likewise, magic squares (*murabba'āt-i alvāḥ-i al-dād*) are described as being even-times-even (*zawj al-zawj*), even-times-odd (*zawj al-fard*) or odd-times-odd (*fard al-fard*), [making them similar to geomantic figures], this by way of example. Furthermore, when someone performs a geomantic operation to inquire into the feasibility of a goal, for instance, and learns that it will take considerable time to accomplish what they desire (or that they will not be able to do so at all), and they wish to bring it to pass more quickly, they can then resort to letter magic (*tadbir*) [by way of a magic square]. This procedure involves selecting a divine Name or an invocation (*du'ā'i*) that seems appropriate to the case and whose active property (*khāṣṣiyyat*) is said to bring about the desired effect, applying gematria (*ḥisāb-i jummal*) to the number of letters in that Name or

¹³⁸ He is perhaps to be identified with the celebrated poet Luṭf Allāh b. Sulaymān Shāh Nishāpūrī (d. 812/1409), a Khurasan-based encomiast to the Sarbidars and Timurids known for his ascetic character and expertise in astrology; see *Divān-i Luṭf Allāh Nishābūrī*, Rasūl Ja'fariyān, ed., Tehran: Asnād-i Majlis-i Shūrā-yi Islāmī, 1390 Sh./2011, 117–118; Dawlatshāh Samarqandī, *Tadbkirat al-Shu'arā'*, E.G. Browne, ed., Tehran: Asāṭir, 1382 Sh./2003, 317–325).

¹³⁹ The earliest surviving copy of this treatise (of some 15 in Iran) dates to 1020/1611; an earlier copy is preserved in Hamburg (MS 226/2, copied 971/1563; Storey, *Persian Literature*, vol. 2.3, 481, no. 847). Luṭf Allāh Nishāpūrī Samarqandī was also the author of *Khavāṣṣ-i Ḥurūf u Amāl-i Ashkāl-i Ramlī*, evidently similar in theme (or simply an extract from the *Khulāṣat al-Baḥrayn*), as well as *Nūr al-Raml*, wherein he identifies himself as the student of Zayn al-Ābidin b. Qāsim Iṣfahānī (Rammāl-i Iṣfahānī) and 'Abd al-Ghanī Ḥāfiẓ Shīrvānī (see e.g. Yazd, MS Kāzimaynī 62/1 ff. 1b–96b). Tellingly, a copy of his teacher Shīrvānī's *Anwār al-Raml* is bound together with an anonymous treatise on magic squares, *Kayfiyyat-i A'māl-i A'dād-i Vafq*, in MS Mar'ashi 7142.

¹⁴⁰ I.e., geomantic figures are either even or odd; there is one figure with four dots and one with eight, and the rest have either five, six or seven dots.

invocation, then populating a square (*murabbaʿi*) with the result; if they then carry [the resulting talisman] on their person, they will soon achieve their aim.¹⁴¹

In other words, geomancy tells you the future, and if you don't like what you hear, you can always change the future with letter magic. This prospect, very simply, explains the attractiveness of this and other occult sciences to ruling elites throughout the early modern Islamicate world and their heavy patronage of the same; used in conjunction with lettrism in particular, geomancy promises the direct control of one's political and material fate.¹⁴²

I have shown elsewhere that lettrism was reformulated during this same early 9th/15th-century moment as the primary vehicle of Islamicate neopythagoreanism and philosophical occultism, and advanced as a superior alternative to both mainstream peripatetic-illuminationist metaphysics and sufism.¹⁴³ Significantly, the network of thinkers who accomplished this reformulation appear to have called themselves Ikhwān al-Ṣafā' in conscious reference to the shadowy group of 4th/10th-century neoplatonist-neopythagorean encyclopedists of the same name. By the early 9th/15th century this network included most prominently Shaykh Badr al-Dīn of Simavna (d. btw. 819–823/1416–1420) and ʿAbd al-Raḥmān al-Bisṭāmī (d. 858/1454) in Anatolia¹⁴⁴ and Ṣāʿin al-Dīn Turka Iṣfahānī (d. 835/1432) and Sharaf al-Dīn ʿAlī Yazdī (d. 858/1454) in western Iran, with Sayyid Ḥusayn Akhlāṭī (d. 799/1397), famed Tabrizi Kurdish occultist and physician at the Mamluk court in Cairo, standing as the network's pivot.¹⁴⁵ Ibn Turka, in turn, stands as its leading theoretician in the east, being responsible for the systematization of this philo-

¹⁴¹ MS Majlis 12563/1, pp. 2-3. Interestingly enough, in the preface above the author lists ʿUmar Khayyām (d. 526/1132) as an authority on geomancy and/or magic squares (MS Majlis 12563/1, p. 2). Cf. *Majmaʿ al-Uṣūl*, an anonymous work that similarly combines geomancy and lettrism (MS Dānishgāh 2306).

¹⁴² As Maḥmūd Kāshī, *munajjim* to the Timurid Iskandar Sulṭān (r. 812–817/1409–1414), argues for to the necessity of astrology: "The wise soul manages the actions of the heavens, just as the farmer manages the power of nature by ploughing and watering"; Laurence P. Elwell-Sutton, "A Royal Timurid Nativity Book", in: *Logos Islamikos: Studia Islamica in Honorem Georgii Michaelis Wickens*, Roger M. Savory and Dionisius A. Agius, eds., Toronto: Pontifical Institute of Mediaeval Studies, 1984, 119–136, here: 130. On the need for a functionalist approach to Islamicate occultism see Dimitri Gutas, "Certainty, Doubt, Error: Comments on the Epistemological Foundations of Medieval Arabic Science", in: *Early Science and Medicine* 7/3 (2002), 276–289, here: 278–279.

¹⁴³ See Matthew Melvin-Koushki, *The Quest for a Universal Science: The Occult Philosophy of Ṣāʿin al-Dīn Turka Iṣfahānī (1369–1432) and Intellectual Millenarianism in Early Timurid Iran*, PhD dissertation, Yale Univ., 2012; Melvin-Koushki, "Powers of One".

¹⁴⁴ İhsan Fazlıoğlu, "İlk Dönem Osmanlı İlim ve Kültür Hayatında İhvanı's-Safā ve Abdurrahman Bisṭāmī", in: *Dîvân: İlmi Araştırmalar* 1/2 (1996), 229–240; Fleischer, "Ancient Wisdom and New Sciences", 235; see Melvin-Koushki, "The Quest", 16–19.

¹⁴⁵ See now İ. Evrim Binbaş, *Intellectual Networks in Timurid Iran: Sharaf al-Dīn ʿAlī Yazdī and the Islamicate Republic of Letters*, Cambridge: Cambridge Univ. Press, 2016; Melvin-Koushki, "The Quest", 45–57.

sophical lettrism¹⁴⁶ – the basis of his student and friend Yazdī's historical and occult-scientific outlook. That this intellectual confraternity's interests were not confined to lettrism is suggested by the fact that another of Akhlāṭī's disciples, Ḥasan Abarqūhī, was known strictly as a geomancer, as we saw above. (It will be also recalled that Akhlāṭī himself has been credited as author of the seminal geomantic manual *R. Surkhāb*.)

Unlike the other geomantic authorities surveyed here, Yazdī, a respected mathematician who achieved fame primarily as dynastic historian to the Timurids,¹⁴⁷ produced no manual on the subject. Preserved among his *munsba'āt*, however, is a tract that mounts a strong defense of the science on a number of fronts and appears to be conceived of as a rebuttal to his older colleague Ibn Khaldūn's tendentious and fideistic anti-occultist arguments in the *Muqaddima*.¹⁴⁸ In particular, Yazdī offers two proofs for the validity of geomancy and its pedigree that he clearly holds to be unassailable; these are, of course, lettrist proofs.¹⁴⁹

The correspondence (*tawāfuq*) between the number of triplicities (*muthallathāt*) that are possible in a geomantic operation and the numerical value of the [divine] name *Nūr* (Light),¹⁵⁰ [i.e., 256], constitutes an intuitive indication as to the benefit and goal of the science of geomancy, that being to obtain information about unknown circumstances and bring clarity to ambiguous or cryptic situations by means of specific methods; and perception comes about solely through the overflowing, knowledge-bearing effulgence of the holy name Light. That is, just as light is the cause and means of the manifestation of things, so too does geomancy constitute a means of lifting the obscuring veils from hidden matters. The fact that the number of geomantic figures multiplied by itself¹⁵¹ as well as the number of these figures multiplied by the number of houses [in a geomantic tableau]¹⁵² are both equivalent to the value [of the name *Nūr*] further reinforces this point.¹⁵³

¹⁴⁶ His *K. al-Mafāḥiṣ* in particular stands as the first *summa* of lettrist neopythagoreanism; see Melvin-Koushki, "The Quest", 330–378.

¹⁴⁷ On Yazdī as mathematician see Mahdi Farhani Monfared, "Sharaf al-Din 'Ali Yazdī: Historian and Mathematician", in: *Iranian Studies* 41/4 (2008), 537–547. It is highly significant in the present context that toward the end of his career he appears to have worked at the Samarkand Observatory (Binbaş, *Intellectual Networks*, 65).

¹⁴⁸ For a study of this Ibn Khaldūn-Yazdī debate and a full translation of Yazdī's tract on geomancy see Matthew Melvin-Koushki, "In Defense of Geomancy: Šaraf al-Din Yazdī Rebutts Ibn Ḥaldūn's Critique of the Occult Sciences", in: *Islamicate Occultism: New Perspectives*, Matthew Melvin-Koushki and Noah D. Gardiner, eds., double issue of *Arabica* 64/3 (2017), 346–403.

¹⁴⁹ As Yazdī asserts in his *Nikāt-i Kalimāt al-Tawḥīd*, lettrist proofs are among those immediately and instinctively grasped and admit of no doubt (*az yaqīniyyāt-i fiṭri-st va dar ān hič šakk u šubḥa nist*) (MS Ayasofya 1801, f. 38b).

¹⁵⁰ This value is $50 + 6 + 200 = 256$; there are four triplicities in a geomantic tableau (a triplicity being two Mothers or two Daughters from the first row and the one Niece they produce), so $16 \times 4 \times 4 = 256$.

¹⁵¹ I.e., $16 \times 16 = 256$. The actual phrase here is *murabba'-i 'adad-i ashkāl-i raml*.

¹⁵² Also 16, including that of the Reconciler.

¹⁵³ "Faṣḥi dar Bāb-i Raml", in: *Sharaf al-Din 'Ali Yazdī, Munsba'āt*, Īraj Afshār and Muḥammad Riḍā Abū'ī Mahrizi, eds., Tehran: Thurayyā, 1388 Sh./2009, 87–91, here: 89.

He further defends the traditionalist proof, scornfully dismissed by Ibn Khaldūn, which posits the prophet Daniel as the inventor of geomancy.¹⁵⁴ Here again his primary argument in support of his interpretation of the relevant hadith is a lettrist one: geomancy bears a numerological correspondence to the name Daniel, so the association holds.

We must therefore apply ourselves to this art with such purity of intention (*ṣafā-yi niyyat*), unity of purpose (*vaḥdat-i qaṣd*) and total concentration (*jamʿ-i himmat*) that the soul may thereby achieve a type of correspondence (*munāsabat*) with the all-emanating Source (*mabdaʿ-i fayyāḍ*) that can call forth its effulgence and evoke hidden things to appear as they truly are and as plainly as can be shown. Those who quaff from the flagons of intuition (*kuʿūs-i adbrvāq*) will find their insight confirmed by the fact that the number of dots in all the geomantic figures, [i.e., 96], is equivalent to the value of the name Daniel (Dāniyāl) (God bless and keep him and our own Prophet).¹⁵⁵ From this equivalency based on the method of the secrets of names and numbers it may be understood that the application of this art can only be effective when it occupies the totality of one's attention in singularity of purpose, all centered on the operation to the extent possible, for this science is entirely predicated on mentation and imagination (*pindāri*).¹⁵⁶

As a divinatory methodology based on number theory, then, geomancy fit neatly into the newly resurgent neopythagorean-neoplatonic (*fiṭhāghūrī-aflāṭūnī*) project that was sweeping the early 9th/15th-century intellectual scene, and served as the primary foundation for the philosophical-scientific and spiritual-political adventurousness that characterized the early modern Persianate world; this conceptual confluence is plainly reflected in Luṭf Allāh Nishāpūrī's innovative marriage of geomancy and letter magic on the one hand and Yazdī's pointedly lettrist defense of the science on the other.

So far geomancy's scientific classification; but what of its primary mechanism? Yazdī alludes to the answer in the last passage above. Geomancy, like all occult sciences as practiced in the early modern Islamo-Christianate world, is predicated on the hermetic principle of correspondence (*munāsaba*): as above, so below. This union of supernal and infernal, of spirit and matter, of eternity and time is governed by the law of (gendered) attraction or love (*ḥubb*, *maḥabba*, *ʿishq*) obtaining between all bodies, from planetary to atomic.¹⁵⁷ The neoplatonists hold

¹⁵⁴ *Muqaddima*, 108.

¹⁵⁵ Dāniyāl is numerically equivalent to 96 (4 + 1 + 50 + 10 + 1 + 30).

¹⁵⁶ "Faṣḥi," 89–90.

¹⁵⁷ Empedoclean love, fused with the erotic motion of Plotinus; see Sajjad Rizvi, *Mullā Ṣadrā and Metaphysics: Modulation of Being*, London: Routledge, 2009, 85. Cf. the *Iḳbrwān al-Ṣafāʿ*'s 37th epistle on love ('Divine wisdom and solicitude has bound all existents together with a single bond and strung them together in a single arrangement'; 'God is the First Beloved, and the celestial sphere revolves solely out of longing (*shawq*) for Him and love (*maḥabba*) for permanence'), Ibn Sinā's *Risāla fi l-Ishq* and Ibn ʿArabī's embrace of the sufi concept of *al-nikāḥ al-sāri fi jamīʿ al-dharāri* ('the marriage act pervading all atoms'), referred to by Ibn Turka as 'perpetual bonding' (*al-ʿaql al-sāʿir*) and 'pervasive coupling' (*al-nikāḥ al-sāri*) (Melvin-Koushki, "The Quest", 364). Most famously the basis for Isaac Newton's theory of gravitation, this doctrine was ubiquitous in early modern Islamo-Christian thought,

that these two principles erect a hierarchy of being that may be ascended and descended at will; the neopythagoreans hold that number is the only shuttle capable of returning the human intellect to its transcendent, unitary origin. Properly manipulated, then, the primordial virtue of number may be activated in a way that grants knowledge of the unseen – past, present and future, essences and secret thoughts. Geomancy is one such method of number manipulation whose efficacy depends on the correspondence of above to below along the neoplatonic Chain of Being.¹⁵⁸

This point is explicitly stated by no less a figure than Shams al-Din Muḥammad Khafri, prominent astronomer and philosopher under the early Safavids; significantly for our purposes here, George Saliba has called Khafri the most important planetary theorist of the 10th/16th century, and believes him to be the first thinker to redefine mathematics “as a descriptive language of scientific processes,” foreshadowing “the modern conception of the role of mathematics in science in general.”¹⁵⁹ (At this point it is perhaps unsurprising to learn that Khafri was father of the most important lettrist author of 10th/16th-century Iran, Maḥmūd Dihdār Shīrāzī (fl. 984/1576), Shaykh Bahāʾī’s (d. 1030/1621) teacher in the occult sciences.)¹⁶⁰ In a simply titled *R. dar Raml*, one of two Persian treatises he wrote on geomancy, Khafri succinctly explains the philosophical-cosmological framework of

from the political philosophy of Jalāl al-Din Davānī in the 9th/15th century to the millenarian empiricism of Francis Bacon in the early 11th/17th. As the latter memorably describes it in his essay on the subject, “Cupid; or the Atom” (*Of the Wisdom of the Ancients*, in *The Works of Francis Bacon*, James Spedding et al., eds., 14 vols., Boston: Brown & Taggard 1860–1864, vol. 6, 729–730): “The fable [of Cupid, or Love,] relates to the cradle and infancy of nature, and pierces deep. This Love I understand to be the appetite or instinct of primal matter; or to speak more plainly, the natural motion of the atom; which is indeed the original and unique force that constitutes and fashions all things out of matter ... [T]he summary law of nature, that impulse of desire impressed by God upon the primary particles of matter which makes them come together, and which by repetition and multiplication produces all the variety of nature, is a thing which mortal thought may glance at, but can hardly take in.”

¹⁵⁸ On the Chain of Being theory generally see Arthur Lovejoy, *The Great Chain of Being: A Study of the History of an Idea*, Cambridge, Mass.: Harvard Univ. Press, 1936; on the same in Islamicate thought see Aziz Al-Azmeh, *Arabic Thought and Islamic Societies*, London: Croom Helm, 1986, 2–13.

¹⁵⁹ “Kafri, Šams-al-Din,” in: *Encyclopaedia Iranica*; see also his “A Sixteenth-Century Arabic Critique of Ptolemaic Astronomy: The Work of Shams al-Din al-Khafri”, in: *Journal for the History of Astronomy* 25 (1995), 13–38. While Saliba’s celebration of Khafri is expressly and problematically teleological and positivist, I rely on it here to emphasize the Shirazi thinker’s neopythagorean bent – wholly elided in Saliba’s account; see Melvin-Koushki, “Powers of One”. On Khafri’s position in the school of Shiraz see Reza Pourjavady, *Philosophy in Early Safavid Iran: Najm al-Din Maḥmūd al-Nayrizi and His Writings*, Leiden: Brill, 2011, 37–40 and passim; on his philosophical thought see Firouzeh Saatchian, *Gottes Wesen – Gottes Wirken: Ontologie und Kosmologie im Denken von Šams-al-Din Muḥammad al-Ḥafri* (gest. 942/1535), Berlin: Schwarz, 2011.

¹⁶⁰ See Matthew Melvin-Koushki, “Maḥmūd Dehdār Šīrāzī”, in: *Encyclopaedia Iranica*; and Melvin-Koushki, *The Occult Science of Empire*.

the science (here with reference to the method Hidāyat Allāh terms the Geomantic East):¹⁶¹

Inasmuch as the subject of this science is the dots and the 16 figures [they form which] populate 16 cells (*buḡūtāt*), it is necessary to explain the nature and associations (*mansūbāt*) of each. The methodology (*vaḍ'*) of this science is based on correspondence (*munāsabat*); thus because the world of generation and corruption (*'ālam-i kawn u fasād*) is composed of four elements (*'anāšir*), four single dots (*nuḡṭa-yi fard*) are first arranged like so, ∴, with each standing for a given element. The first dot is called fire (*ātash*), the second air (*bād*), the third water (*āb*), the fourth earth (*khāk*). And because the natures (*tabāyī'*) of these elements are each composed of two qualities (*kayfiyyat*),¹⁶² each of these four dots may be doubled like so, ∷. This figure is known as Populus (*jamā'at*), while the first is known as Via (*tariq*). Because Via is the simplest of the figures it is called the Universal Intellect (*'aql-i kull*), and because Populus features all possible dots it is called the Universal Soul (*naḡs-i kull*). Because the eight [composite] qualities of natures are lower than the four elements and the elements lower than the [Universal] Intellect which occupies the second degree (*martaba*) after God Most High, who occupies the first degree of oneness (*yaki*), it is appropriate (*munāsib*) that fire be represented by a single dot, air by two dots, water by four dots and earth by eight dots. This arrangement is also referred to with the letters *ABDH*,¹⁶³ with *A* assigned to the first dot, *B* to the second, *D* to the third and *H* to the fourth ... And because the cells are 16 in number, or four times four according to the number of natures, and the sum of the above numbers is 15, it is appropriate that the Universal Intellect be assigned to the fifteenth cell, with the sixteenth cell being assigned to the Universal Soul, the closest of contingent beings (*mumkināt*) to the Universal Intellect immediately above it.¹⁶⁴

Most importantly for history of science, this neopythagorean-neoplatonic framework underlying all occultist practice is itself predicated on the principle of secondary causality vertically cascading down the Chain of Being. Here Yazdi takes to task his anti-occultist, fideist interlocutor, Ibn Khaldūn, who heaps scorn and censure on those who investigate secondary causes,¹⁶⁵ by counterposing geomancy as a science precisely devoted to such investigation:

Now all those who rashly deny the excellence of this noble art appear to do so because they hold to the belief that knowledge of the unseen (*'ilm-i ghayb*) was only vouchsafed

¹⁶¹ The *nisba* Khafri is often misread as Khidri by catalogers, e.g. in Storey, *Persian Literature*, vol. 2.3, 487, no. 856 (40). At least 20 MS copies survive of Khafri's *R. dar Raml* in Iran alone, a testament to its popularity; he also authored a second short geomantic treatise entitled *Davāzdah Maṭla'* that cites the *Sā'ir al-Awliyā* and 'Abd Allāh Balyāni (i.e., certainly the *Jahān al-Raml*; see e.g. MS Majlis 12509/1, ff. 1b–5a; this *majmū'a* also includes Nūr al-Dīn Abhari's commentary on Tūsi's *Vāfi* and a commentary on Ṭabasi's *Aḥkām*).

¹⁶² Viz., hot or cold and wet or dry.

¹⁶³ I.e., 1-2-4-8.

¹⁶⁴ *R. dar Raml*, MS Majlis 3931/3, pp. 128–129, MS Majlis 15755, ff. 37b–38a, Princeton MS New Series 1177/2, ff. 18b–19a. This helpful if modest-sized treatise (10 folios or 27 pages respectively) is notable for its theoretical and methodical approach to the subject.

¹⁶⁵ *Muqaddima*, 429–431. Yet Ibn Khaldūn cannot simply be dismissed as an Ash'ari occasionalist; it is painfully ironic, to say the least, that he inveighs against the investigation of secondary causes in a masterwork devoted precisely to investigating the causes behind the rise and fall of civilizations (my thanks to Justin Stearns for this observation).

the holy [Prophet], knower of things hidden, and it is impossible for others to attain the same. But it is evident that the geomancer does not in any way claim to have knowledge of the unseen. The most that he claims in this regard is that his art is governed by a number of general principles and rules that were initially discovered through revelation and inspiration (*vahy u ilhām*), then collected and taught from one generation to the next down to the present day, and that it is possible through knowledge of these principles and rules and total concentration (*ṣidq-i andisha*) on the form (*sūrat*) the unseen returns (*radd*) to gain information about how specific situations will unfold at specific times and locations.¹⁶⁶

This is no perversion of belief; to the contrary, it is a clear proof (*dalil*, *burhān*) that the order of existents (*nizām-i mawjūdāt*) in the Chain of Being (*tartib-i silsila-yi mukavvanāt*) necessarily manifests according to the eternal knowledge and will and wisdom of God. If, hypothetically speaking, it were not so, and events simply happened randomly (*kayfa mā ttafaqa*) or on the basis of temporary, corruptible principles, then one could never accurately deduce from the dots and dashes produced by the geomancer's stylus (*kill*) the manner of unfolding of past, future and present events. The truth of this statement will be obvious to the intelligent who know the basis on which geomantic readings are derived.¹⁶⁷

In short: If geomancy can't work, no other natural or mathematical science can. But since geomancy does work, is testable and true, we can know – and control – the cosmos. Nor should it be considered a religiously suspect science; after all, the idea that a binary code underlies and shapes physical reality has long been a staple among logocentric metaphysicians, Muslim, Jewish and Christian alike – a host of whom embraced the Danielic art.¹⁶⁸

Geomancy's claim in the Arabo-Persian tradition to be an *experientially verifiable* (*mujarrab*)¹⁶⁹ and fundamentally *rational*¹⁷⁰ mathematico-natural science, fea-

¹⁶⁶ Cf. the famous dictum of the Pythagorean Alcmaeon (fl. 500–450 BCE) in reference to artificial divination (tr. in Robert Drew Hicks, *Diogenes Laertius*, Cambridge, Mass.: Harvard Univ. Press, 1995, 397): “Of things invisible, as of mortal things, only the gods have certain knowledge; but to us, as men, only inference from evidence is possible.”

¹⁶⁷ Yazdī, “Faṣḥi,” 87–88.

¹⁶⁸ On Judeo-Arabic geomancy see Blanca Villuendas Sabaté, *Judaico-Islamica. La geomancia en los manuscritos judeo-árabes de la Gueniza de El Cairo*, Cordoba: Cordoba Univ. Press, 2015.

¹⁶⁹ *Tajriba*, experiential assaying, became a major theme in Islamic occultism during the 8th–10th/14th–16th centuries in particular; geomancers, lettrists, astrologers and alchemists were increasingly emphatic about the *mujarrab* nature of their findings and procedures. As ‘Alī Ṣafī Kāshifī (d. 939/1533), for example, son of the great Timurid polymath and occultist Ḥusayn Vā‘iz Kāshifī, asserts in his lettrist treatise *Hirz al-Amān min Fitān al-Zamān*, which cites both Akhlāfī and Ibn Turka: ‘Experiential assaying aimed at manifesting the effects and properties of the letters has often been successfully repeated; and experience is among the forms of proof that produce certainty (*tajriba yaki az ithbāt-i yaqīn ast*)’, (see Melvin-Koushki, “The Quest,” 279). This, it must be emphasized, represents a significant departure from aristotelian precedent, wherein *empeiria* is a mode of general knowledge considerably inferior to *epistēmē*, ‘science’ (my thanks to Nicholas Harris for this observation). As that may be, the empirical nature of occult science was already asserted by such authorities as Abū Ma‘shar Balkhī (d. 272/886), father of conjunction astrology, who influentially declared his discipline to be a science founded on analogical reasoning (*qiyās*) and experience (*tajriba*), the latter entailing observation (*mu‘āyana*) and direct discovery (*al-*

turing falsifiable truth claims, must therefore be taken far more seriously by historians of science than it has been to date. That such a massive chapter in that tradition's development – the 13th–19th centuries, when it was at the apex of its sophistication and popularity throughout Afro-Eurasia as a primary application of the *quadrivium* – has been left wholly unwritten, while its Arabo-Latin sister has been recuperated as a worthy, if still marginal, object of historiographical investigation, testifies to the *reductionist-colonialist double standard* that still structurally defines history of science as a field.

A positivist case in point is Francis Bacon (d. 1626), roundly hailed as the father of experimental science, and hence precursor of “scientific modernity,” despite his profoundly occultist, perennialist and millenarian outlook. Indeed, some scholars, following Frances Yates (d. 1981), now predicate his revolutionary natural-philosophical project on precisely that outlook. That is, Bacon's occultism did not negate his experientialist rationalism, but drove it.¹⁷¹ Most significantly for our purposes here: he elevated *artificial divination* – considered far inferior to natural divination from Plato onward – to the status of an experimental natural science, pointedly renaming it natural divination to this end.¹⁷² Now geomancy, of course, was with astrology the most popular and technically sophisticated form of artificial divination in the early modern Islamo-Christianate world. And as we have seen, early modern Muslim geomancers asserted much the same experimental status for the science; they too valorized artificial divination over natural.¹⁷³ More, and un-

ʿunjūd al-ḥādīr) (Charles Burnett, “The Certitude of Astrology: The Scientific Methodology of al-Qabīṣī and Abū Maʿshar”, in: *Early Science and Medicine* 7/3 (2002), 198–213, here: 210, 212). An identical argument is made for medicine by Ibn Hindū (d. 410/1019 or 420/1029), who rather uses the terms *raṣād* and *mushābada* for observation and personal inspection respectively (Gutas, “Certainty, Doubt, Error”, 279–280).

¹⁷⁰ I use this descriptor here in a nontechnical sense to signify the philosophically coherent outlook of our occultists. But its technical sense also applies: rational demonstration (*burhān ʿaqlī*) was a key, if contested, element of occult-scientific theory, together with experience, observation and analogical reasoning, especially with respect to astrology. Most notably, al-Kindi (d. ca. 260/873) himself allowed the application of *burhānī* status to predictive sciences, although his student Abū Maʿshar disagreed (Burnett, “The Certitude”, 212).

¹⁷¹ See e.g. Stephen A. McKnight, “The Wisdom of the Ancients and Francis Bacon's New Atlantis”, in: Allen G. Debus and Michael T. Walton, eds., *Reading the Book of Nature: The Other Side of the Scientific Revolution*, Kirksville, Mo.: Sixteenth Century Journal Publishers, 1998, 91–109.

¹⁷² A. P. Langman, “The Future Now: Chance, Time and Natural Divination in the Thought of Francis Bacon”, in: Andrea Brady and Emily Butterworth, eds., *The Uses of the Future in Early Modern Europe*, New York: Routledge, 2010, 142–158, here: 152.

¹⁷³ It should be noted that this formal distinction between natural and artificial divination, as codified by Cicero (d. 43 BCE) in his *De divinatione*, is not as systematically made in the Arabo-Persian tradition. Most famously, however, Ibn Khaldūn – a committed if confused anti-occultist – distinguished between *kāhins*, who engage in possession divination, and *ʿarrāfs*, who engage in inductive (*taḥakkum*) divination (Melvin-Koushki, “In Defense of Geomancy”). The Tunisian historian's blatant, fideist bias against and ignorance of Islamicate intellectual history aside, the pointed reclassification of geomancy as both a mathematical and an Islamic science in the Persianate world was designed to retire this binary.

like their Christian counterparts, they formally elevated geomancy from the natural to the mathematical sciences in recognition of its remarkable ability to empirically demonstrate the validity of neopythagorean cosmology. For if number is the uncreated, all-creative matrix of the cosmos, then the disciplined manipulation of number – in this case even-odd bits – can be expected to give reliable data about that cosmos. Islamo-Christianate geomancy thus exemplifies, in its formal classifications and practical deployments, that early modern brand of “Physico-Mathematicall-Experimentall Learning”¹⁷⁴ which has been so thoroughly investigated and feted in the case of Europe alone.

Islamicate geomancy, in sum, a mainstream and vital *Western* science, represents a uniquely strategic object of investigation in its role as epitome of early modern Western intellection (much of it heavily neopythagorean): its study cannot but neatly undercut the prevailing teleological eurocentrism of the field. But those historians of science wishing to rehabilitate such instances of early modern Islamicate high occultism as expressly rational, experientialist pursuits must navigate between the Scylla and Charybdis erected by post-Enlightenment positivism and the 20th-century reactions thereto. On the one hand, they must refrain from reflexively disappearing mind into matter in modern *materialist-realist* fashion; in a mindless cosmos “occult science” is a flat impossibility, an epistemological travesty, a contradiction in terms. On the other, they must forswear the equally extremist *religionist* approach counterposed by the great antimodernist-Islamicist Henry Corbin (d. 1978), which strives to disappear occult science into the immaterial – and hence apolitical – dumping ground that is mysticism, is religion.¹⁷⁵ (C. G. Jung’s (d. 1961) *psychologization* of occultism initially represented a third if equally ahistorical way, and powerfully restored it as a category to public and scholarly consciousness; but this tack was totalized by his immediate intellectual heirs, then largely commandeered for religionist ends.¹⁷⁶) The paradoxical logic of binaries

¹⁷⁴ Thus John Wilkins’ summarization, in 1660, of the nascent Royal Society’s mission (Peter Dear, *Discipline and Experience: The Mathematical Way in the Scientific Revolution*, Chicago: Chicago Univ. Press, 1995, 2). Needless to say, I use “experimental” here strictly in its early modern sense.

¹⁷⁵ Wouter J. Hanegraaff, *Esotericism and the Academy: Rejected Knowledge in Western Culture*, Cambridge: Cambridge Univ. Press, 2012, 299–302.

¹⁷⁶ Hanegraaff, *Esotericism*, 281–295; on the Jungian concept of *synchronicity* as the engine of all divination see n. 11 above. The temptation to simply fold all forms of divination into the catchall category of *intuition* – the hunch, the gut feeling – must therefore be resisted, for it effectively consigns the study of complex divinatory sciences like geomancy to psychology and anthropology. The most recent example of such a psychologizing approach is that of Peter Struck, who in a programmatic article on the cognitive history of divination in ancient Greece advances an otherwise compelling argument for reconceptualizing the mantic arts as a natural and hence perennial expression of what he terms cognitive intuition (“2013 Arthur O. Lovejoy Lecture: A Cognitive History of Divination in Ancient Greece”, in: *Journal of the History of Ideas* 77/1 (2016), 1–25); this summarizes his larger argument in *Divination and Human Nature: A Cognitive History of Intuition in Classical Antiquity*, Princeton: Princeton Univ. Press, 2016. This is not to deny the central role of intuition,

being what it is, of course, these violently opposed approaches have successfully pursued the same goal: the rendering of “postclassical” Islamicate philosophy a pale, passive, otherworldly, strictly unscientific and utterly un-Western pursuit.

But the examples of Tūsi, Nishāpūri, Yazdi, Khafri and Hidāyat Allāh Shīrāzi presented above explode historiographical positivism and religionism alike. As they conclusively show, the elite Muslim (occult) philosopher-scientists of the post-Mongol Persianate world professionally deployed their highly technical, experientialist forms of divination and magic to very *material* ends as a *rational* application of their panpsychist-monist neopythagorean cosmology. And chief among these ends: the occultist construction and maintenance of henological West-East *empire*.

Conclusion

The ‘science of sand,’ as a quintessentially Arabic and thoroughly Islamicized system of divination and precise cognate to the *I Ching*, developed from obscure origins to become, by the 7th/13th century, a mature and mainstream mathematical-natural occult science of wide appeal throughout the Islamo-Christian world. Predicated on a neopythagorean-neoplatonic system and animated by the twin principles of correspondence and secondary causation, geomancy was of particular and enduring interest to thinkers and rulers throughout the Persianate world during the 7th–11th/13th–17th-century moment most especially. Its great philosophical-scientific virtue lay in its ability to empirically support the neopythagorean cosmology of its scholarly practitioners; its professional virtue lay in its status as magnet for royal patronage. For Mughal, Safavid, Ottoman and other elites, the science became part and parcel of the performance of sainthood and sacral kingship because it promises granular, prophetic-grade insight into, and thus power over, history. That is to say, such sciences of divination, as the very term connotes, were crucial to the quest for *divinization* pursued by many Turko-Mongol Perso-Islamic sovereigns and other messianic claimants in the run-up to the Islamic millennium.¹⁷⁷

Nor is geomancy a strictly premodern fad; it and its offspring remain in common use to the present. Though charlatans may again abound in these latter, darkling, post-millennial days, this to the point that *rammāli*, incompetently done, is now an imprisonable offense in Iran, there is no longer any reason for scholars to deny geomancy and its fellow mathematical occult sciences their rightful place in Islamicate intellectual and imperial history, or for historians of science to deem it a

or *dhawq*, in Islamicate theories of divination; after all, Yazdi asserts geomancy to be entirely predicated on *mentation*. But such mentation must involve the processing of intuitions by means of intellect (‘*aql*), and that within a rigorous philosophical (neopythagorean-neoplatonic) and experientialist framework.

¹⁷⁷ The English term divination pivots on the Greek conception, from Plato onward, of possession by the divine as sole guarantee of philosophical truth.

“superstitious art” of little philosophical-scientific worth.¹⁷⁸ As Ayatollah Hasan Ḥasanzāda Āmulī (b. 1346/1928) recalls of his famed, departed colleague Ayatollah Ṭabāṭabā’ī (d. 1402/1981):

The esteemed teacher ... was holding a lesson one night when something prompted him to relate the following anecdote: ‘There was a gathering [I attended] at which someone known to have expertise in geomancy was asked to identify a certain person. The geomancer performed a geomantic operation and came up with a name. Those present scoffed that a huge number of people have that name. So he repeated the operation and came up with his father’s name. Those present scoffed that a huge number of people so named have that as their father’s name. So he repeated the operation and came up with his occupation. Those present scoffed that a huge number of people so named and whose father is so named have that as their occupation. Things continued in this vein until the geomancer produced the man’s telephone number. Said those present: “To object at this point really wouldn’t be fair!”¹⁷⁹

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¹⁷⁸ Thus Lynn Thorndike’s (d. 1965) characterization of geomancy in his monumental *A History of Magic and Experimental Science*, 8 vols., New York: Columbia Univ. Press, 1923–1958, which mentions the science often in passing; see e.g. vol. 2, ch. 39.

¹⁷⁹ *Hizār u Yak Kalima*, 6 vols., Qom: Būstān-i Kitāb, 1381 Sh./2002, vol. 5, 411–412. The same scholar also produced a *R. dar Raml* (Qom, n.d.). As Hidāyat Allāh makes clear, ‘Allāma Ṭabāṭabā’ī’s geomancer thus used at least three distinct geomantic cycles to progressively narrow down the identity of the quesited.

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